MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ARITHMETIC LOGIC UNIT, DIGITAL, KEY CODE FBA

THIS SPECIFICATION INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic arithmetic logic unit module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.0 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FBA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/174-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

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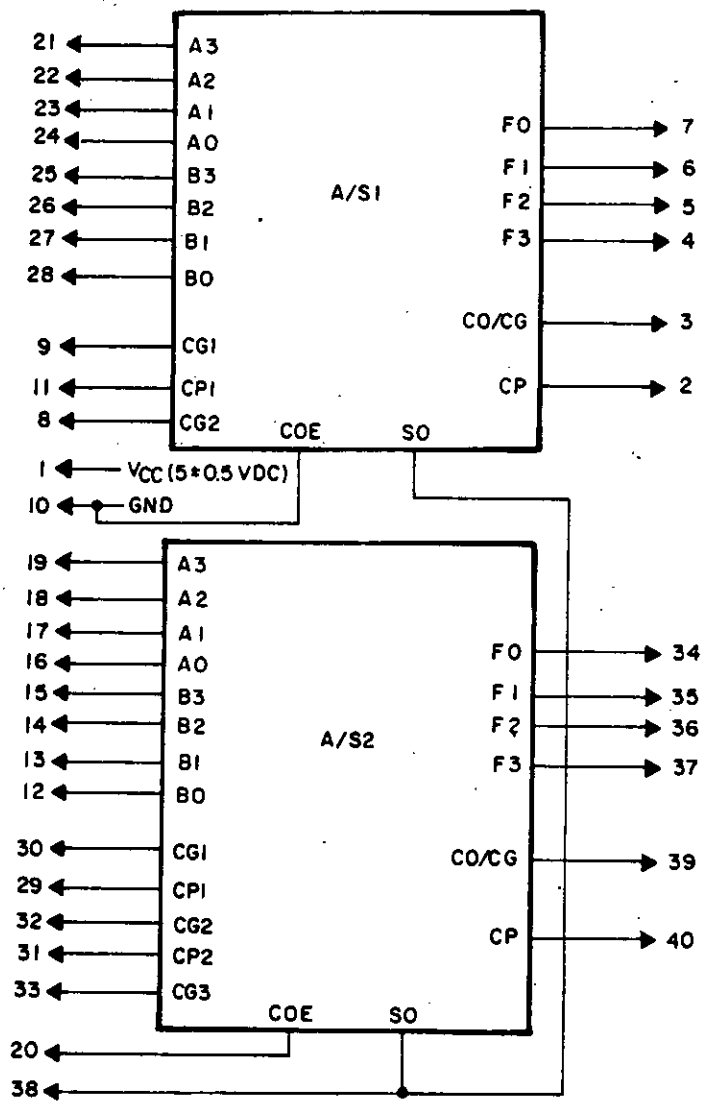


FIGURE 1. FBA module function diagram, arithmetic logic unit.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low-loaded)	III	S	0.0	0.6	0.0	0.6	0.0	0.6	0.0	0.6	V
B1-B25	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
B26	Input breakdown current	IV	S	0	2	0	2	0	2	0	2	mA
C1-C26	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D	Isolation	VI	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
E1-E12	Static output voltage, low level	VII	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
F1-F12	Static output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G1-G18	Input current, low level	IX	X	0.10	1.26	0.10	1.26	0.10	1.26	0.10	1.26	mA
G19-G21	Input current, low level	IX	X	0.05	0.84	0.05	0.84	0.05	0.84	0.05	0.84	mA
G22	Input current, low level	IX	X	0.05	0.63	0.05	0.63	0.05	0.63	0.05	0.63	mA
G23-G26	Input current, low level	IX	X	0.05	0.42	0.05	0.42	0.05	0.42	0.05	0.42	mA
H1-H18	Input current, high level	X	X	0.05	60	0.05	60	0.05	60	0.05	60	µA
H19-H21	Input current, high level	X	X	0.05	40	0.05	40	0.05	40	0.05	40	µA
H22-H26	Input current, high level	X	X	0.05	20	0.05	20	0.05	20	0.05	20	µA
J1-J12	Output short circuit current	XI	S	10	40	10	40	10	40	10	40	mA
K	Power supply current	XII	X	5	74	5	74	5	74	5	74	mA
L1-L12	Transition time (t_{TLH}), C_L = 40 pF	XIII	X	2	45	2	45	2	45	2	45	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L13-L24	Transition time (t_{TLH}), C_L = 140 pF	XIII	Q	4	60	4	60	4	60	4	60	ns
M1-M12	Transition time (t_{THL}), C_L = 40 pF	XIV	S	2	15	2	15	2	15	2	15	ns
M13-M24	Transition time (t_{THL}), C_L = 140 pF	XIV	Q	2	30	2	30	2	30	2	30	ns
N1, N6-N13, N15, N20-N27	Propagation delay time (t_{PLH}), C_L = 40 pF	XV	S	10	101	10	101	10	101	10	101	ns
N2, N16	Propagation delay time (t_{PLH}), C_L = 40 pF	XV	S	12	101	12	101	12	101	12	101	ns
N3, N5, N17, N19	Propagation delay time (t_{PLH}), C_L = 40 pF	XV	S	7	68	7	68	7	68	7	68	ns
N4, N18	Propagation delay time (t_{PLH}), C_L = 40 pF	XV	S	9	65	9	65	9	65	9	65	ns
N14, N28	Propagation delay time (t_{PLH}), C_L = 40 pF	XV	S	10	94	10	94	10	94	10	94	ns
N29-N31	Propagation delay time (t_{PLH}), C_L = 40 pF	XV	S	11	58	11	58	11	58	11	58	ns
N32, N37-N44, N46, N51-N58	Propagation delay time (t_{PLH}), C_L = 140 pF	XV	Q	10	121	10	121	10	121	10	121	ns
N33, N47	Propagation delay time (t_{PLH}), C_L = 140 pF	XV	Q	12	121	12	121	12	121	12	121	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N34, N36, N48, N50	Propagation delay time (t_{PLH}), C_L = 140 pF	XV	Q	10	88	10	88	10	88	10	88	ns
N35, N49	Propagation delay time (t_{PLH}), C_L = 140 pF	XV	Q	10	85	10	85	10	85	10	85	ns
N45, N59	Propagation delay time (t_{PLH}), C_L = 140 pF	XV	Q	10	114	10	114	10	114	10	114	ns
N60-N62	Propagation delay time (t_{PLH}), C_L = 140 pF	XV	Q	11	78	11	78	11	78	11	78	ns
P1, P6-P13, P15, P20-P27	Propagation delay time (t_{PHL}), C_L = 40 pF	XVI	S	10	81	10	81	10	81	10	81	ns
P2, P16	Propagation delay time (t_{PHL}), C_L = 40 pF	XVI	S	12	100	12	100	12	100	12	100	ns
P3, P5, P17, P19	Propagation delay time (t_{PHL}), C_L = 40 pF	XVI	S	7	48	7	48	7	48	7	48	ns
P4, P18	Propagation delay time (t_{PHL}), C_L = 40 pF	XVI	S	9	67	9	67	9	67	9	67	ns
P14, P28	Propagation delay time (t_{PHL}), C_L = 40 pF	XVI	S	10	82	10	82	10	82	10	82	ns
P29-P31	Propagation delay time (t_{PHL}), C_L = 40 pF	XVI	S	7	50	7	50	7	50	7	50	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P32, P37-P44, P46, P51-P58	Propagation delay time (t_{PHL}), C_L = 140 pF	XVI	Q	10	101	10	101	10	101	10	101	ns
P33, P47	Propagation delay time (t_{PHL}), C_L = 140 pF	XVI	Q	12	120	12	120	12	120	12	120	ns
P34, P36, P48, P50	Propagation delay time (t_{PHL}), C_L = 140 pF	XVI	Q	10	68	10	68	10	68	10	68	ns
P35, P49	Propagation delay time (t_{PHL}), C_L = 140 pF	XVI	Q	10	87	10	87	10	87	10	87	ns
P45, P59	Propagation delay time (t_{PHL}), C_L = 140 pF	XVI	Q	10	102	10	102	10	102	10	102	ns
P60-P62	Propagation delay time (t_{PHL}), C_L = 140 pF	XVI	Q	10	70	10	70	10	70	10	70	ns
R1-R8	Output crosstalk, low level	XVII	Q	0	400	0	400	0	400	0	400	mV
S1-S8	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	mV
T	Power supply transient amplitude	XIX	Q	0	300	0	300	0	300	0	300	mV
U	Power supply filter capacitance	XX	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
V1-V26	Input capacitance	XXI	Q	2	15	2	15	2	15	2	15	pF
W1-W25	Functional test	XXII	X	← Compliance with test conditions of table XXII →								

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through U and W shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4. Following the life test and while still at the maximum class temperature, electrical tests E through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through W shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/174-1 supersedes part number M28787/174. The two parts are interchangeable.

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6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	407 mW (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current, mode select input (S0)	
Low level (V_{IL} = 0.0 V)	0.84 mA maximum
(V_{IL} = 0.3 V)	0.8 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	2 mA maximum
Input current, carry propagate (CP1, CP2)	
Low level (V_{IL} = 0.0 V)	0.43 mA maximum
(V_{IL} = 0.3 V)	0.4 mA maximum
High level (V_{IH} = 2.4 V)	20 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input current, carry generate from 1st preceding stage (CG1)	
Low level (V_{IL} = 0.0 V)	1.26 mA maximum
(V_{IL} = 0.3 V)	1.2 mA maximum
High level (V_{IH} = 2.4 V)	60 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input current, carry generate from 2nd preceding stage (CG2)	
Low level (V_{IL} = 0.0 V)	0.84 mA maximum
(V_{IL} = 0.3 V)	0.8 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input current, carry generate from 3rd preceding stage (CG3)	
Low level (V_{IL} = 0.0 V)	0.42 mA maximum
(V_{IL} = 0.3 V)	0.4 mA maximum
High level (V_{IH} = 2.4 V)	20 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input current, carry out enable (COE)	
Low level (V_{IL} = 0.0 V)	0.63 mA maximum
(V_{IL} = 0.3 V)	0.6 mA maximum
High level (V_{IH} = 2.4 V)	20 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input current, operands (A0 through B3)	
Low level (V_{IL} = 0.0 V)	1.26 mA maximum
(V_{IL} = 0.3 V)	1.2 mA maximum
High level (V_{IH} = 2.4 V)	60 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Output voltage, all outputs (F0-F3, CO/CG, CP)	
Low level (I_{OL} = 4.0 mA)	0.3 V maximum
High level (I_{OH} = -0.4 mA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.3 V)	4.0 mA
High level (V_{OH} = 2.4 V)	-0.4 mA

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6.5 Module function. This module contains two adder/subtractor units (A/S1 and A/S2). Each A/S accepts two 4-bit words (A0, A1, A2, A3 and B0, B1, B2, B3) and produces a 4-bit output (F0, F1, F2, F3). The output function is determined by the state on the mode select line (S0). The inputs and outputs of the A/S may be considered to be active level low or active level high. The add and subtract operations are performed on the entire word with carries or borrows propagated between bits of different weight. The arithmetic may be performed in 1's complement or 2's complement. To achieve high speed operation, the A/S is designed to be used in a full internal lookahead system. Full internal lookahead is used inside each A/S unit to propagate carries between bits. External lookahead functions over the 4-bit block are available as outputs. These outputs are labeled CO/CG (carry out/carry generate) and CP (carry propagate) on the logic symbol. The carry in to the device is formed from a set of carry generate and carry propagate inputs so that three A/S's can be interconnected without any additional gates. See figure 2 for an example of interconnections for a 12-bit full lookahead carry adder/subtractor. The pin labeled COE (carry out enable) input controls the CO/CG output. When COE is high, CO/CG becomes a carry out which can be used to ripple carries between blocks. The CG1 input can be used for a ripple carry input, since this signal is sufficient to produce a carry in. The logic equation for COE operation is as follows:

$$(\overline{CG1}) + (\overline{CP1}) \cdot (\overline{CG2}) + (\overline{CP1})\ (\overline{CP2})\ (\overline{CG3}) = C_{in}\ \text{(internal)}$$

$$(\overline{CO/CG} = (\overline{CG}) + (\overline{CP}) \cdot (C_{in}) \cdot (COE)$$

The mode select line (S0) input determines module operation as shown in table II.

TABLE II. Mode select line operation.

Control input (S0)	Operation
	(Active low operands)
L	A SUBTRACT B = F
H	A ADD B = F
	(Active high operands)
L	A SUBTRACT B = F
H	A ADD B = F
	(Active low operands with inverted B)
L	A ADD B = F
H	A SUBTRACT B = F
	(Active high operands with inverted B)
L	A ADD B = F
H	A SUBTRACT B = F

6.5.1 Input to output relationship. Each of the two low-power high-speed A/S units has the input to output relationship of figure 2 except circuit A/S1 does not have the CP2 and CG3 functions available and carry out (COE) is internally enabled for A/S1.

6.5.2 12-bit interconnection. Figure 3 is an interconnection diagram for a 12-bit full look-ahead carry adder/subtractor. In 2's complement arithmetic, a "carry in" must be forced in subtraction; this may be done by connecting the "carry in" to the function control through an inverter. For 1's complement arithmetic, an "end around carry" is required, so the "carry out" should be connected to the "carry in". Care must be exercised in 1's complement mode due to problems associated with logic oscillations between the two possible representations of zero.

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6.5.3 Paralleling of outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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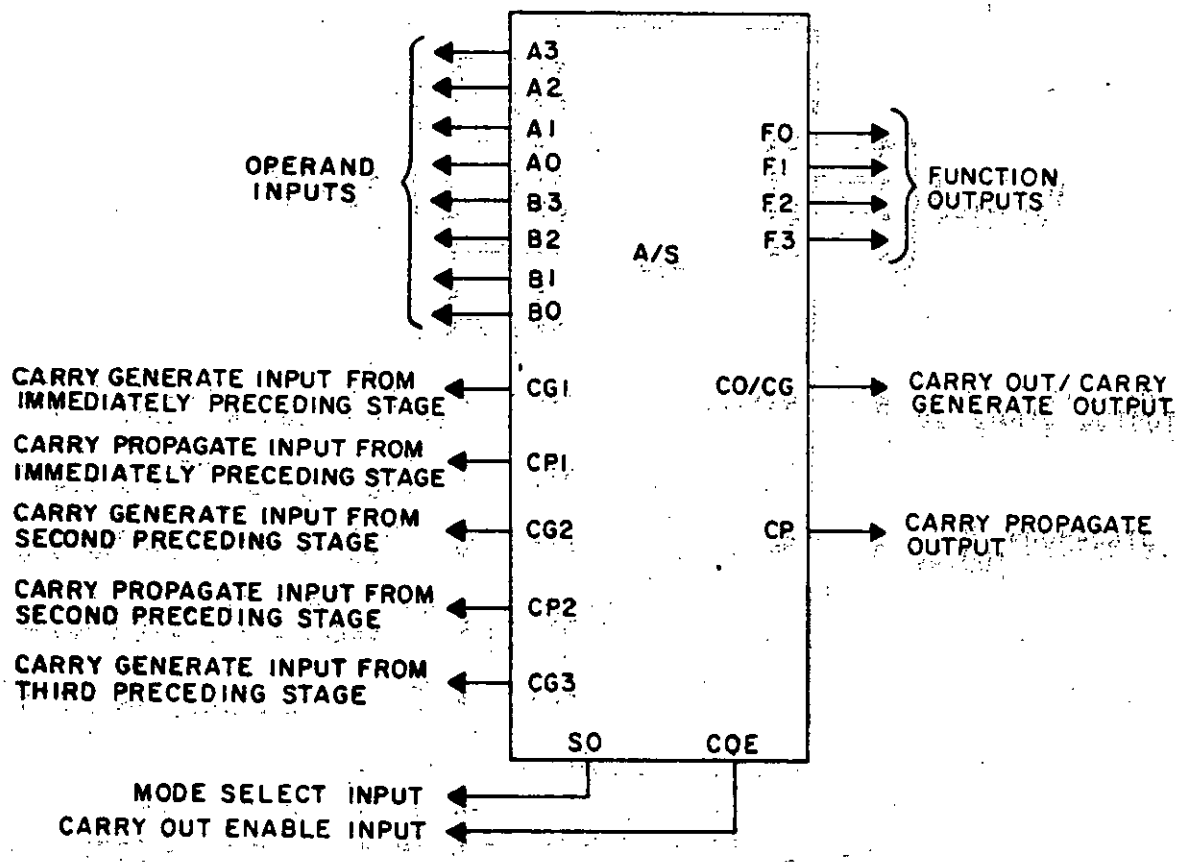


FIGURE 2. Input to output relationship.

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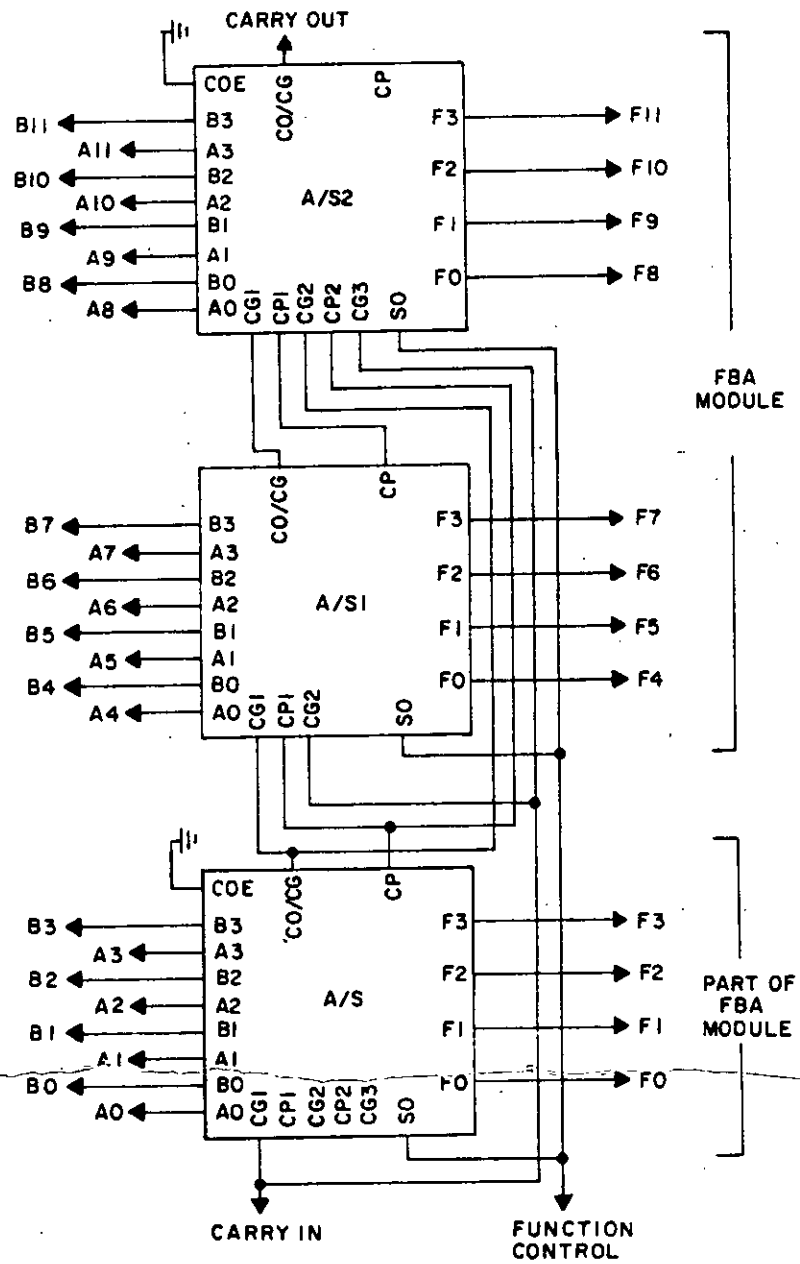
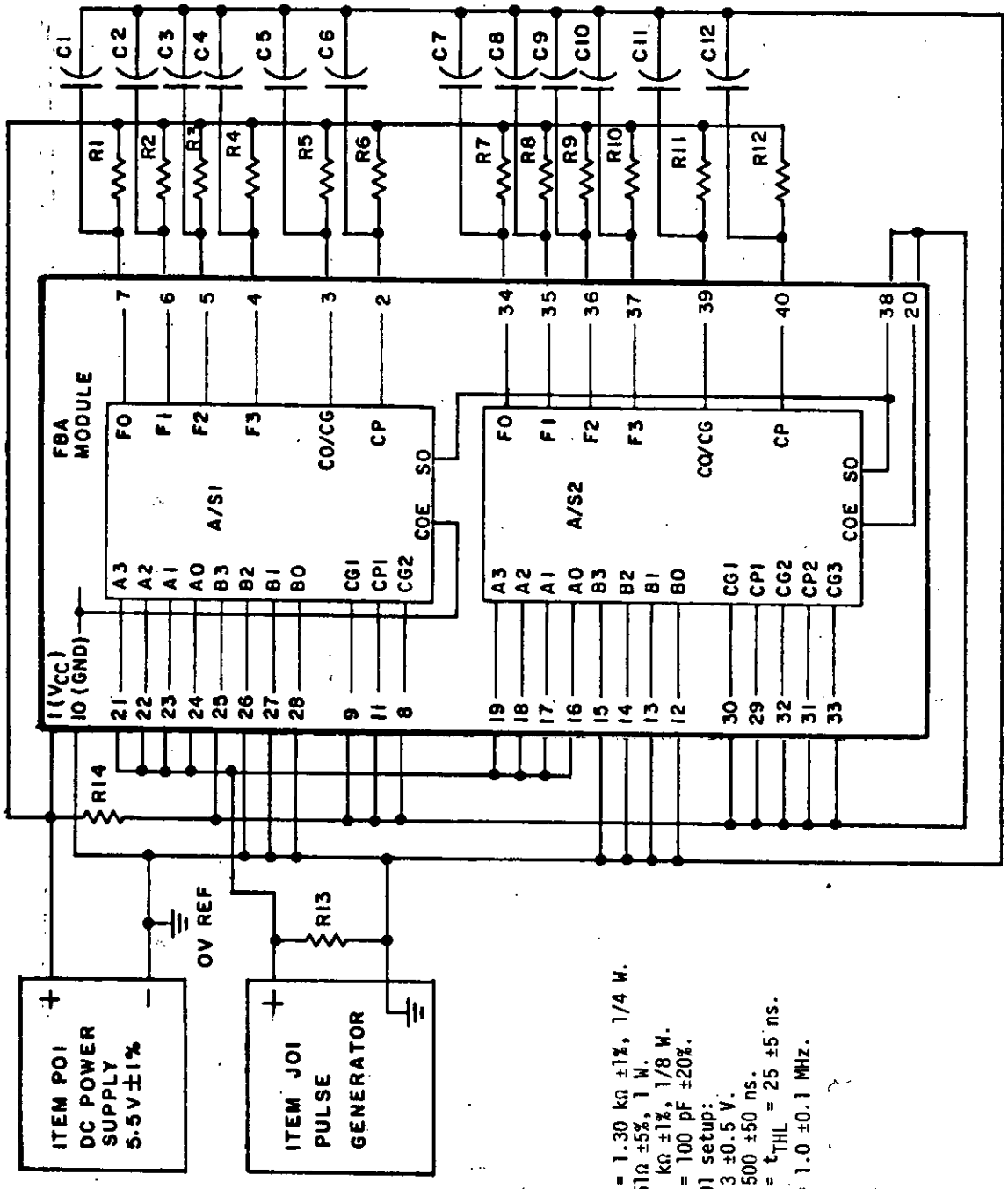


FIGURE 3. 12-bit full look-ahead carry adder/subtractor interconnection diagram.

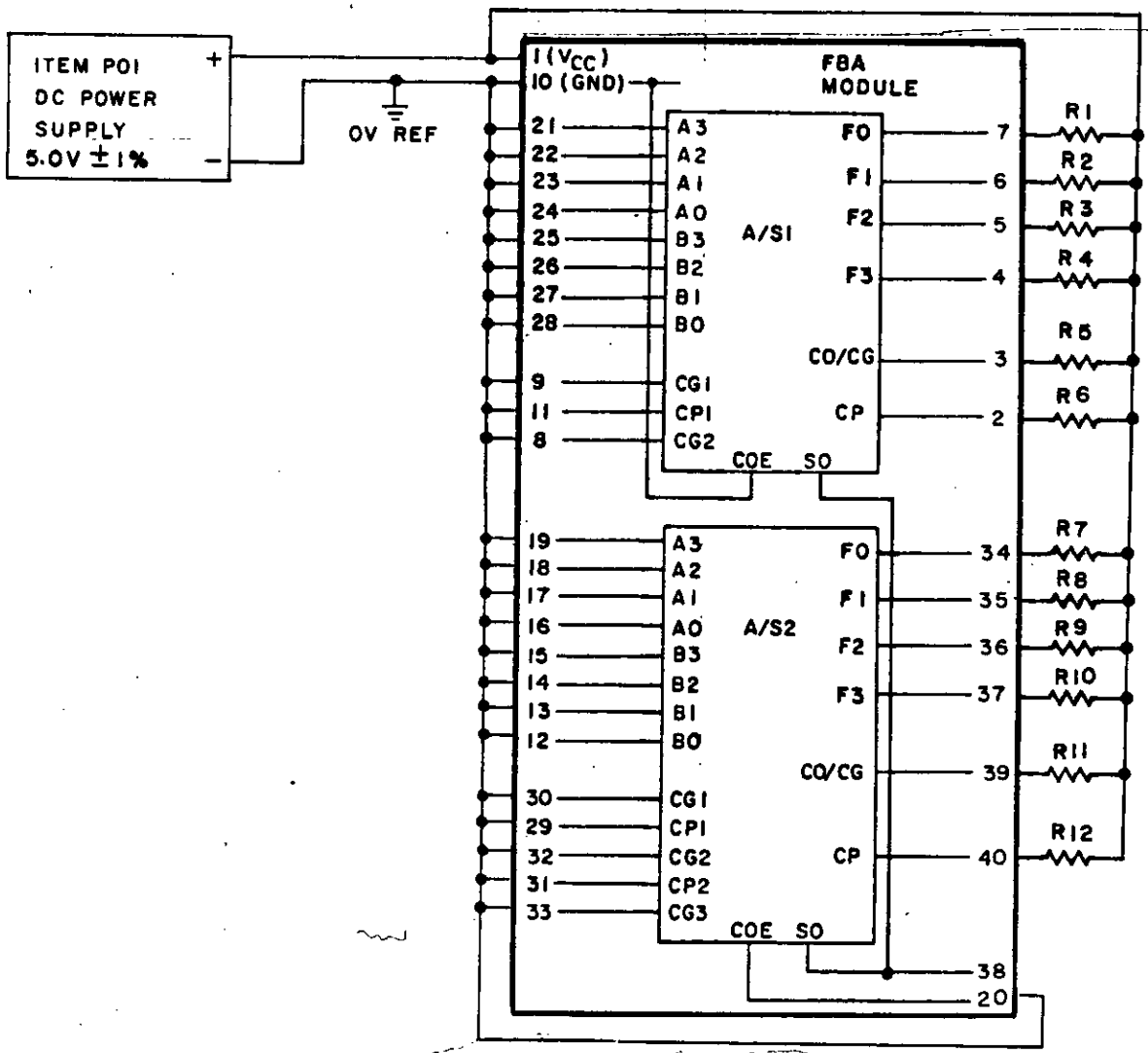
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NOTES:
1. R1-R12 = 1.30 kΩ ±1%, 1/4 W.
2. R13 = 51Ω ±5%, 1 W.
3. R14 = 1 kΩ ±1%, 1/8 W.
4. C1-C12 = 100 pF ±20%.
5. Item J01 setup:
   PA = 3 ±0.5 V.
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   PRR = 1.0 ±0.1 MHz.

FIGURE 4. Operating temperature and life test circuit.

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NOTE: R1-R12 = 1.30 kΩ ±1%, 1/8 W.

FIGURE 5. Component temperature test circuit.

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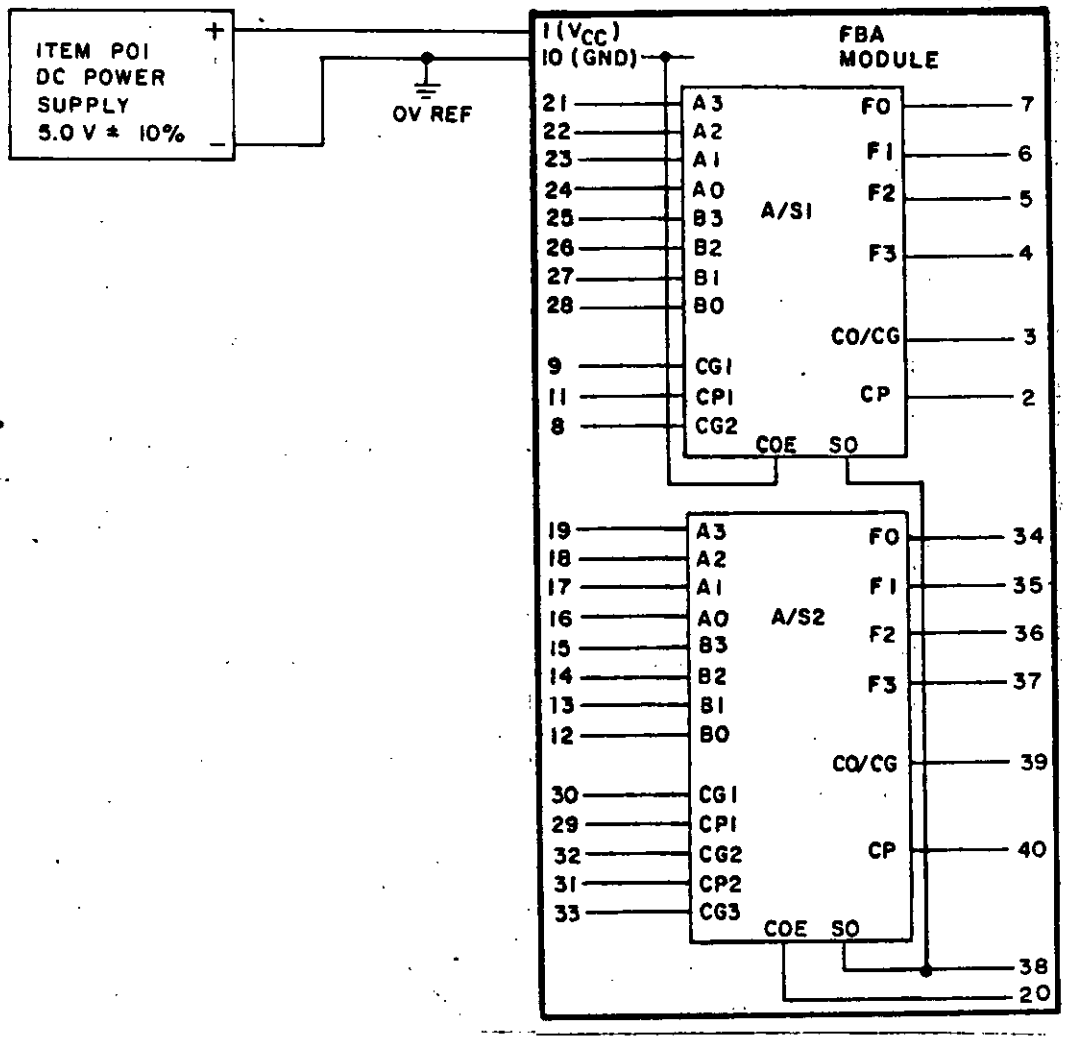


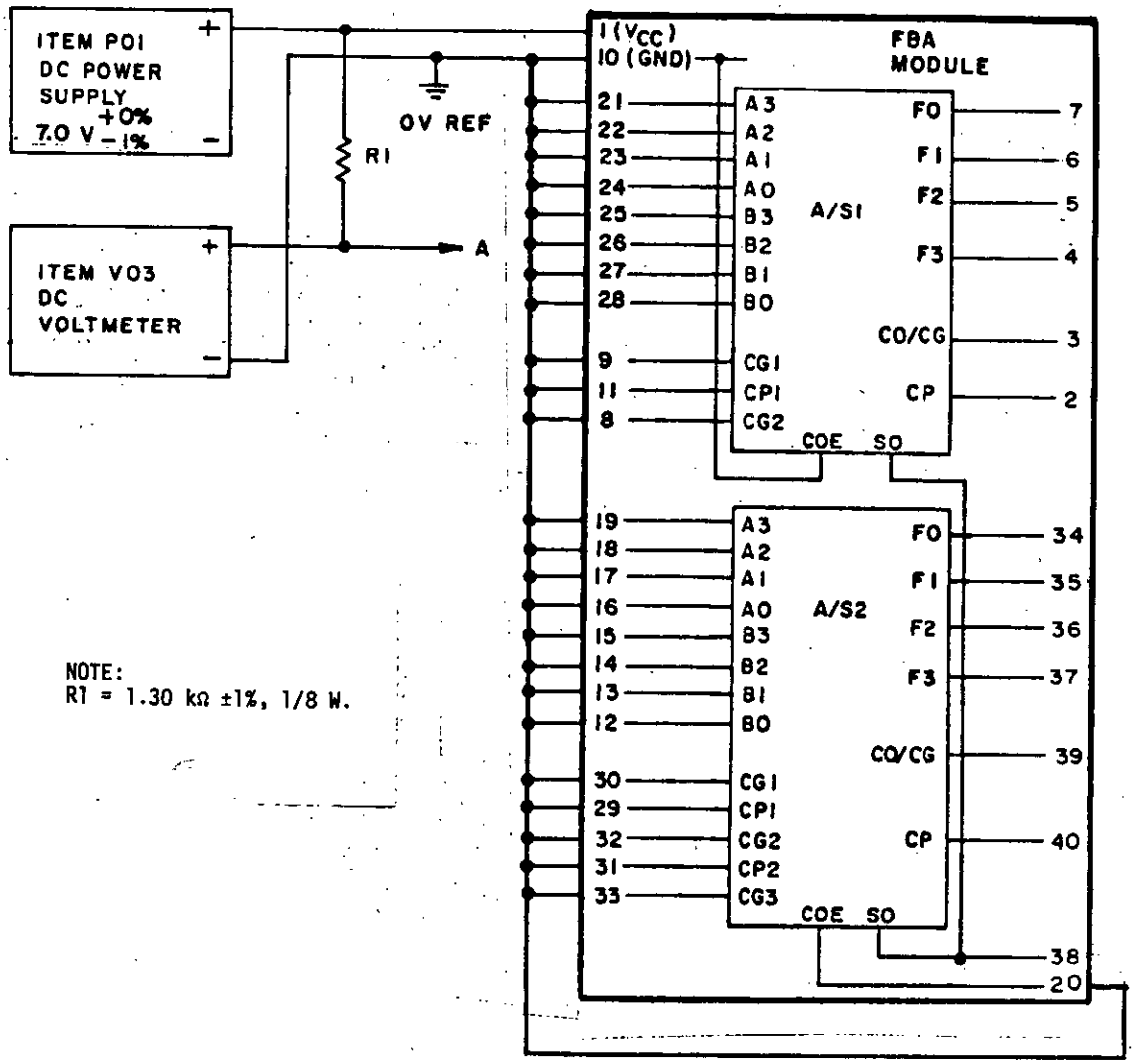
FIGURE 6. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition		Parameter	Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
A1	1F0	V_{OUT} (low loaded)	7	0.0	0.6	V	7
A2	1F1		6				
A3	1F2		5				
A4	1F3		4				
A5	2F0		34				
A6	2F1		35				
A7	2F2		36				
A8	2F3		37				
A9	1C0/CG		3				
A10	2C0/CG		39				
A11	1CP		2				
A12	2CP		40				

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NOTE:
R1 = 1.30 kΩ ±1%, 1/8 W.

FIGURE 7. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

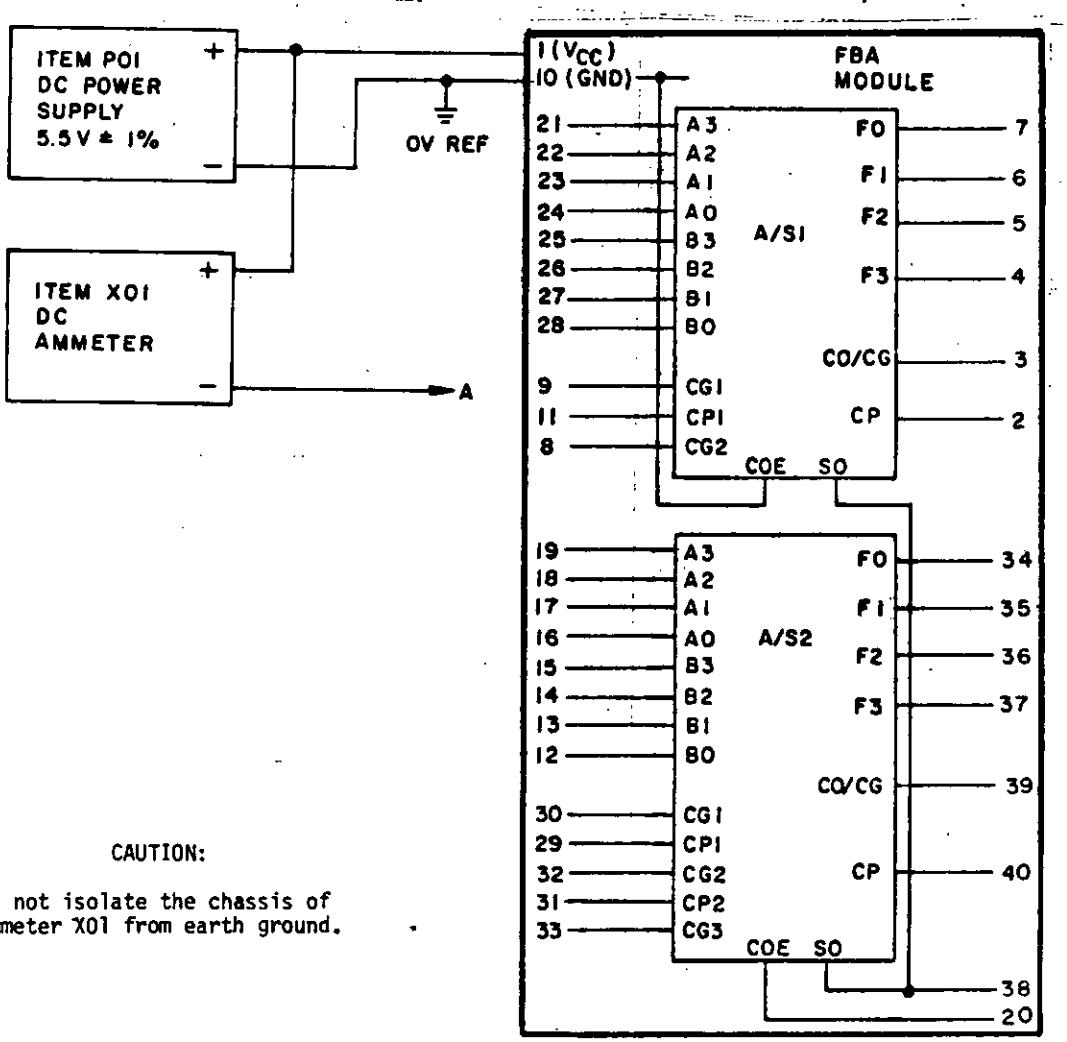
Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
B1	1A3	Input breakdown current	21	0	1	mA	8
B2	1A2		22				
B3	1A1		23				
B4	1A0		24				
B5	1B3		25				
B6	1B2		26				
B7	1B1		27				
B8	1B0		28				
B9	1CG1		9				
B10	1CP1		11				
B11	1CG2		8				
B12	2A3		19				
B13	2A2		18				

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TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
B14	2A1	Input breakdown current	17	0	1	mA	8
B15	2A0		16				
B16	2B3		15				
B17	2B2		14				
B18	2B1		13				
B19	2B0		12				
B20	2COE		20				
B21	2CG1		30				
B22	2CP1		29				
B23	2CG2		32				
B24	2CP2		31				
B25	2CG3		33				
B26	SO		38		2		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 8. Input breakdown current test circuit for table IV.

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TABLE V. Input clamp diode voltage test.

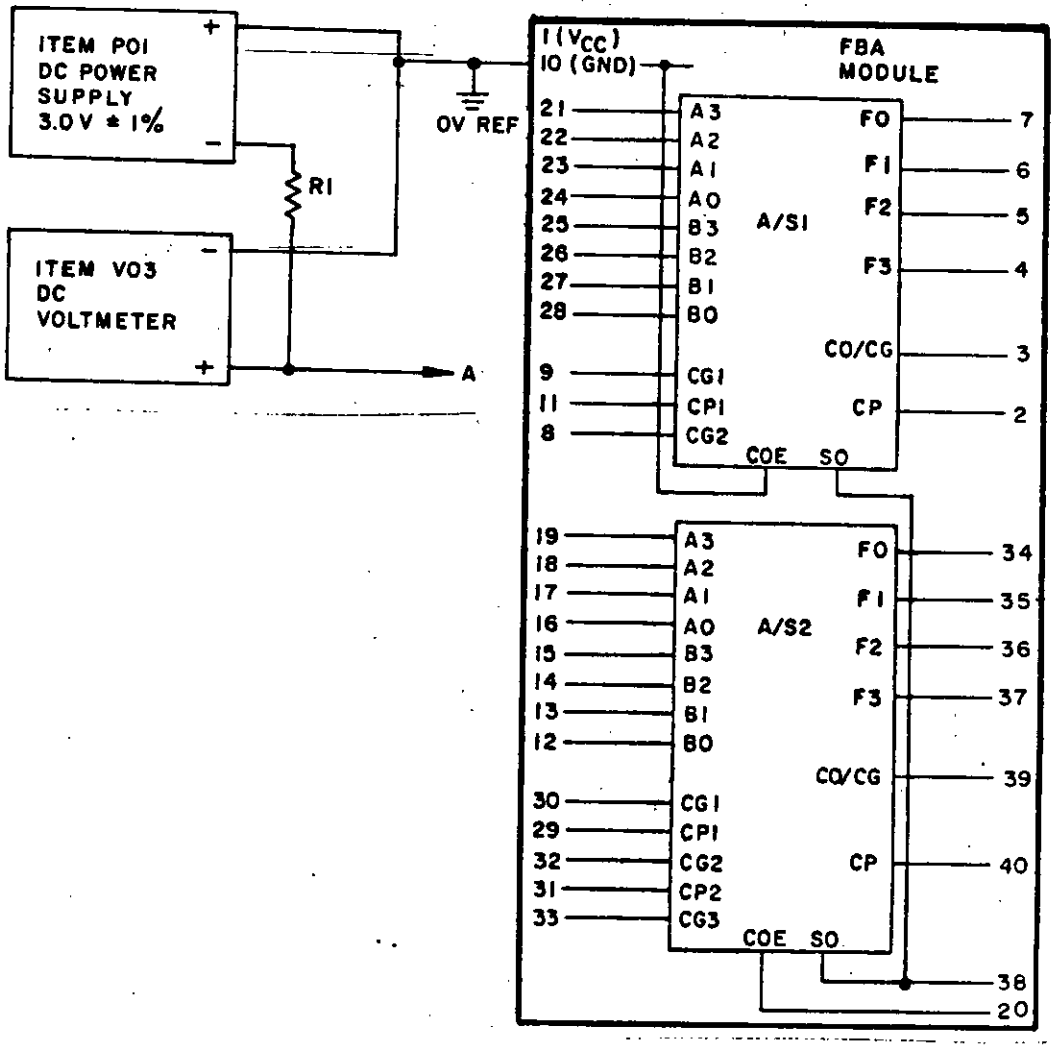
Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
C1	1A3	Input clamp diode voltage	21	-0.4	-1.5	V	9
C2	1A2		22				
C3	1A1		23				
C4	1A0		24				
C5	1B3		25				
C6	1B2		26				
C7	1B1		27				
C8	1B0		28				
C9	1CG1		9				
C10	1CP1		11				
C11	1CG2		8				
C12	2A3		19				
C13	2A2		18				
C14	2A1		17				
C15	2A0		16				
C16	2B3		15				
C17	2B2		14				
C18	2B1		13				

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TABLE V. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
C19	2BO	Input clamp diode voltage	12	-0.4	-1.5	V	9
C20	2COE	↓	20	↓	↓	↓	↓
C21	2CG1		30				
C22	2CP1		29				
C23	2CG2		32				
C24	2CP2		31				
C25	2CG3		33				
C26	SO		38				

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NOTE: R1 = 250Ω ±1%, 1/4 W.

FIGURE 9. Input clamp diode voltage test circuit for table V.

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TABLE VI. Isolation test.

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C ambient		Unit	Test figure
			Min	Max		
D	Isolation	Frame	10	∞	MΩ	10

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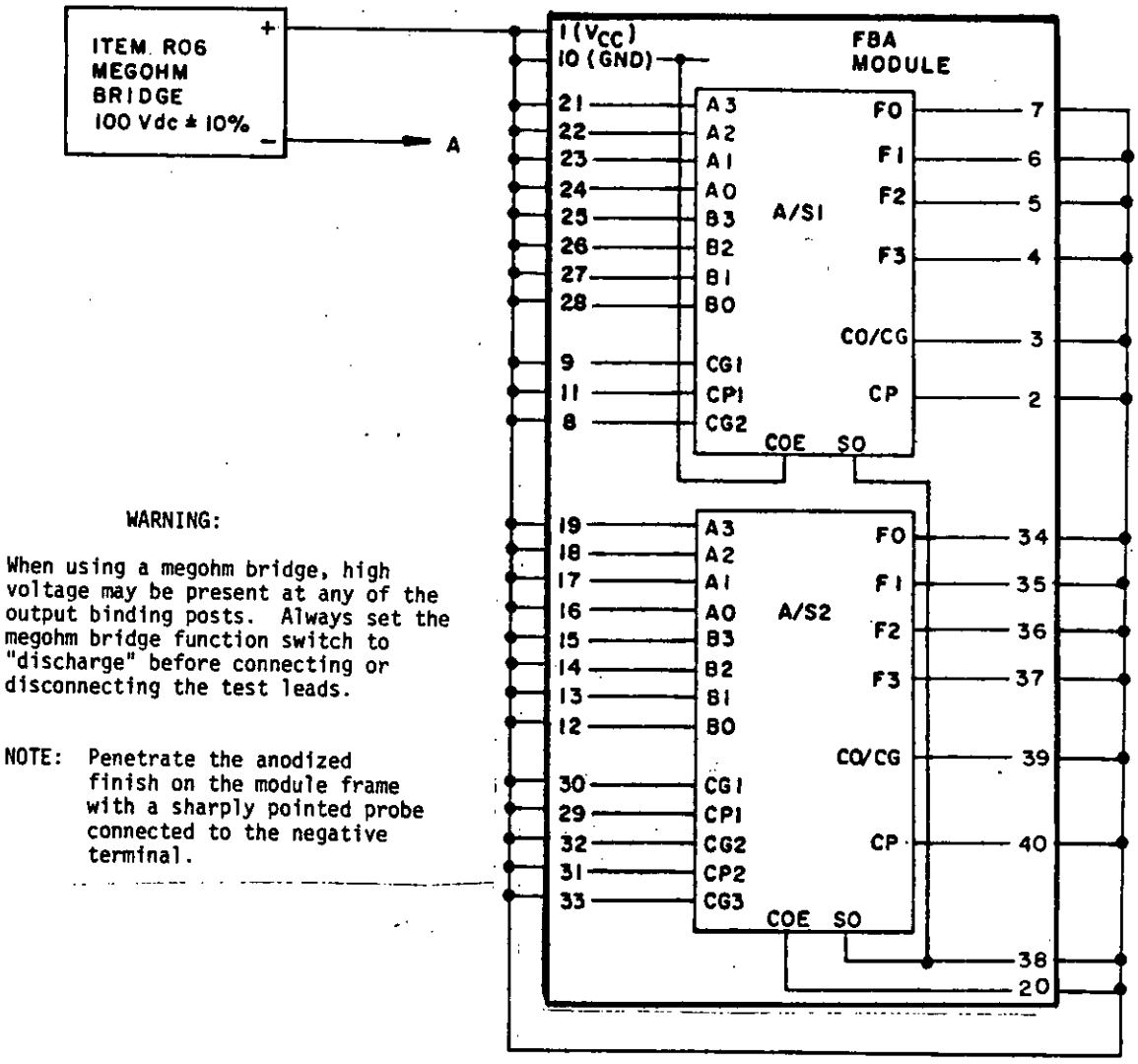


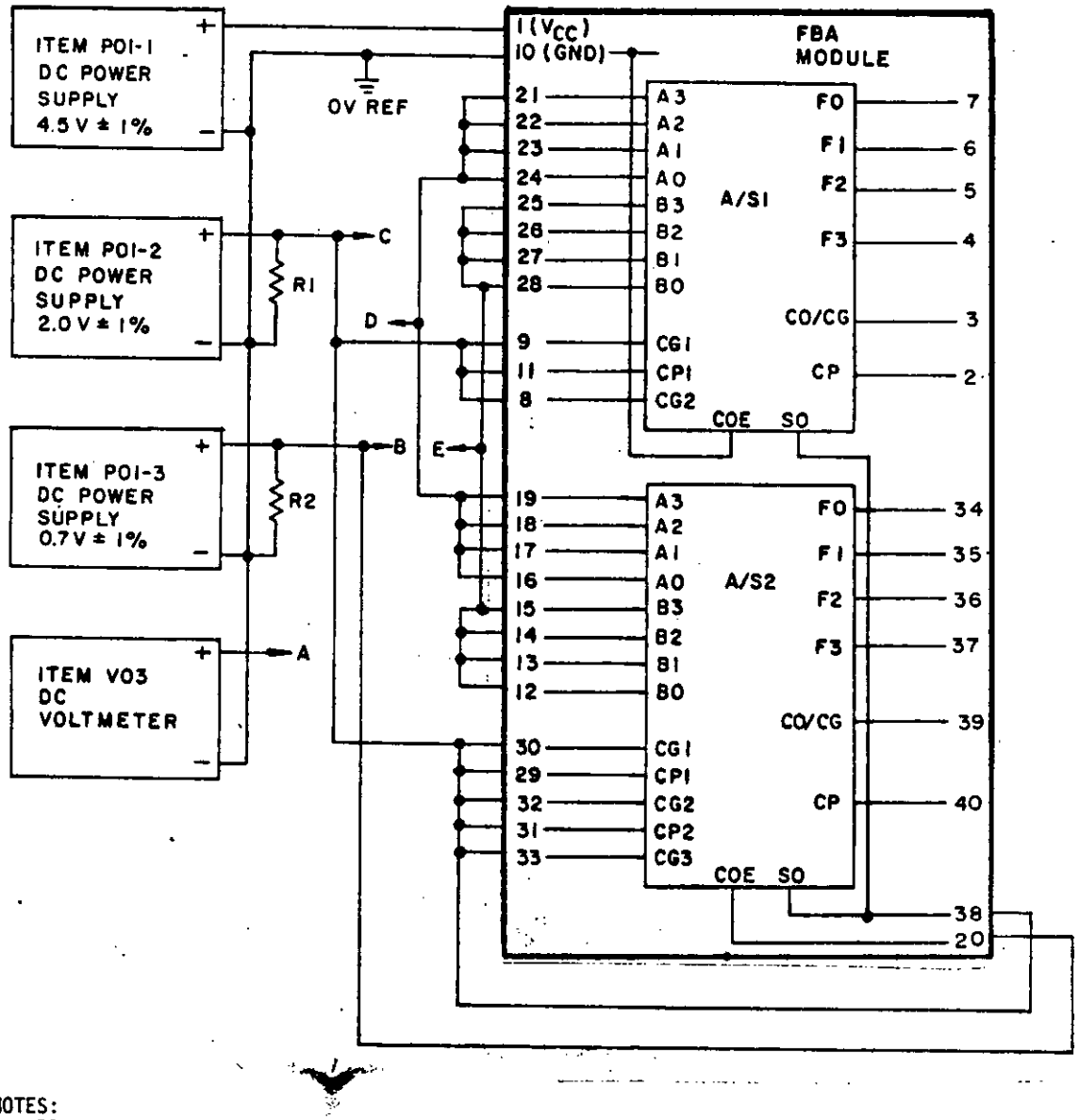
FIGURE 10. Isolation test circuit for table VI.

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TABLE VII. Static output voltage (low-level) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
E1	1F0 Static output voltage, low level	7	E	D	0.0	0.3	V	11
E2	1F1	6						
E3	1F2	5						
E4	1F3	4						
E5	2F0	34						
E6	2F1	35						
E7	2F2	36						
E8	2F3	37						
E9	1CO/CG	3	D, E	Open				
E10	1CP	2						
E11	2CO/CG	39						
E12	2CP	40						

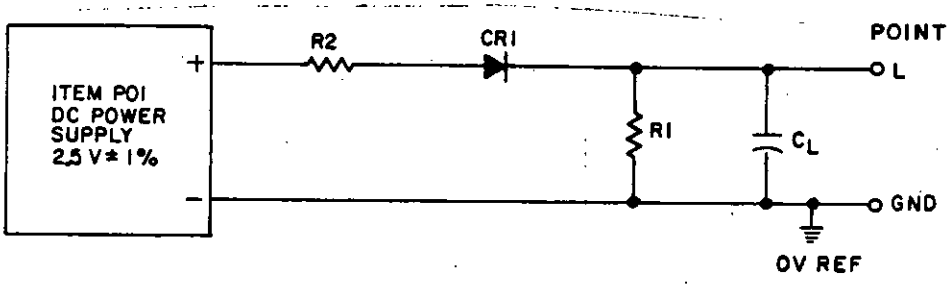
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load all outputs with the load as specified on figure 12.

FIGURE 11. Static output voltage (low-level) test circuit for table VII.

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NOTES:
1. CR1 = 1N916 or equivalent.
2. R1 = 6.04 kΩ ±1%, 1/8 W (current flow of 400 µA when point L is held at 2.4 V).
3. R2 = Select value for current flow of 4 mA ±5% when point L is held at 0.3 V .
4. Test conditions:
   (a) For test conditions L1-L12, M1-M12, N1-N31, and P1-P31, C_L = 40 ±5 pF (including probe capacitance).
   (b) For test conditions L13-L24, M13-M24, N32-N62, and P32-P62, C_L = 140 ±5 pF (including probe capacitance).
   (c) For test conditions R1-R8 and S1-S8, C_L = 50 ±5 pF (including probe capacitance).
5. C_L is not required for dc electrical test conditions E, F, and W.
6. The maximum lead length from point "L" to the appropriate module pin shall be 1 inch (25.4 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

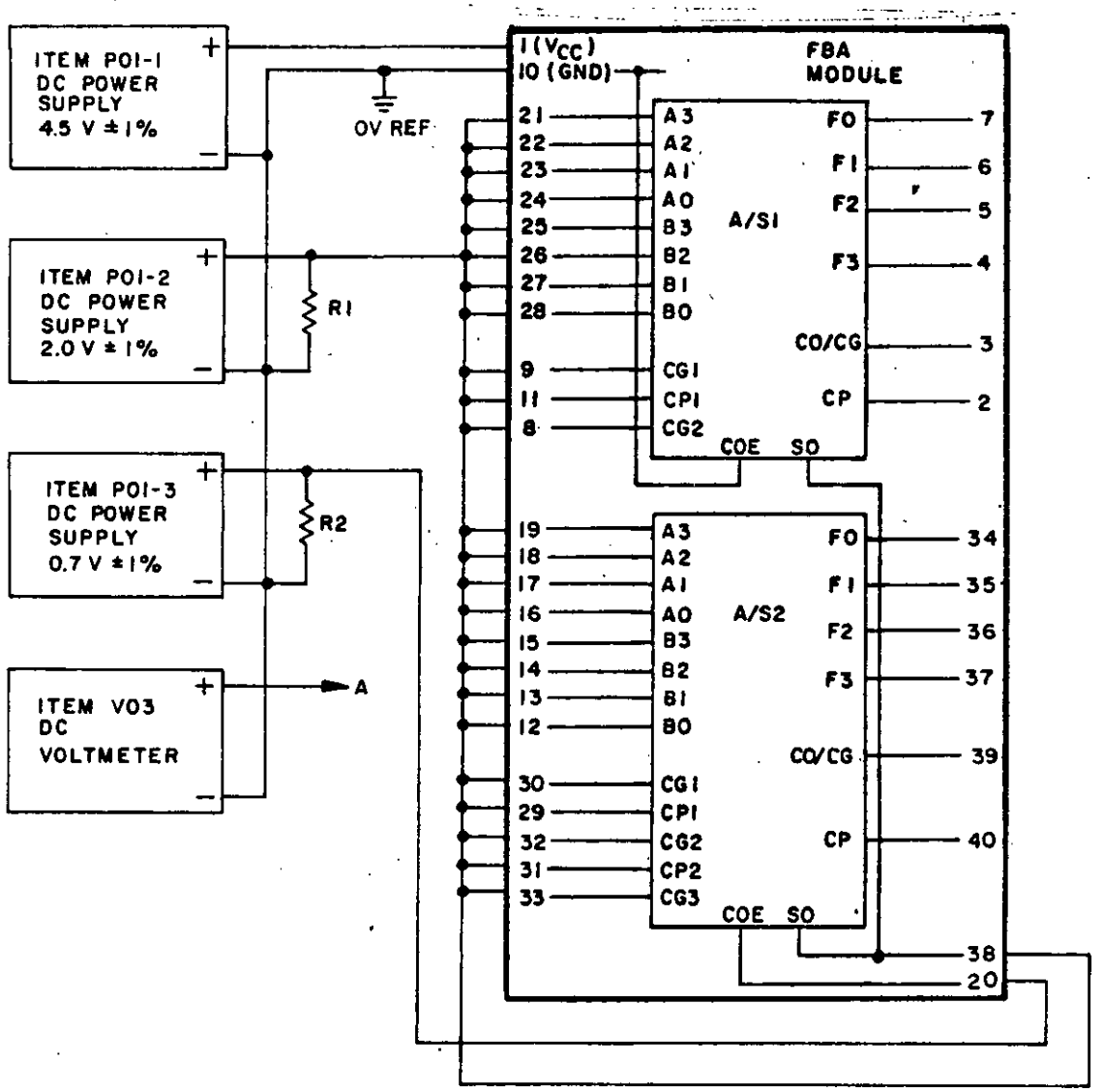
FIGURE 12. Logic load circuit.

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TABLE VIII. Static output voltage (high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
F1	1F0 Static output voltage, high level	7	2.4	4.5	V	13
F2	1F1	6				
F3	1F2	5				
F4	1F3	4				
F5	1C0/CG	3				
F6	1CP	2				
F7	2F0	34				
F8	2F1	35				
F9	2F2	36				
F10	2F3	37				
F11	2C0/CG	39				
F12	2CP	40				

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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load all outputs with load as specified on figure 12.

FIGURE 13. Static output voltage (high-level) test circuit for table VIII.

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TABLE IX. Input current (low-level) test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
G1	1A3	Input current, low level	Open	21	38	0.10	1.26	mA	14
G2	1A2			22					
G3	1A1			23					
G4	1A0			24					
G5	1B3			25					
G6	1B2			26					
G7	1B1			27					
G8	1B0			28					
G9	1CG1			9					
G10	2A3			19					
G11	2A2			18					
G12	2A1			17					
G13	2A0			16					
G14	2B3			15					
G15	2B2			14					
G16	2B1			13					
G17	2B0			12					
G18	2CG1			30					

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TABLE IX. Input current (low-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
G19	1CG2	Input current, low level	Open	8	38	0.05	0.84	mA	14
G20	2CG2			32	38				
G21	SO			38	Open				
G22	2COE			20	38		0.63		
G23	1CP1			11			0.42		
G24	2CP1			29					
G25	2CP2			31					
G26	2CG3			33					

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TABLE X. Input current (high-level) test.

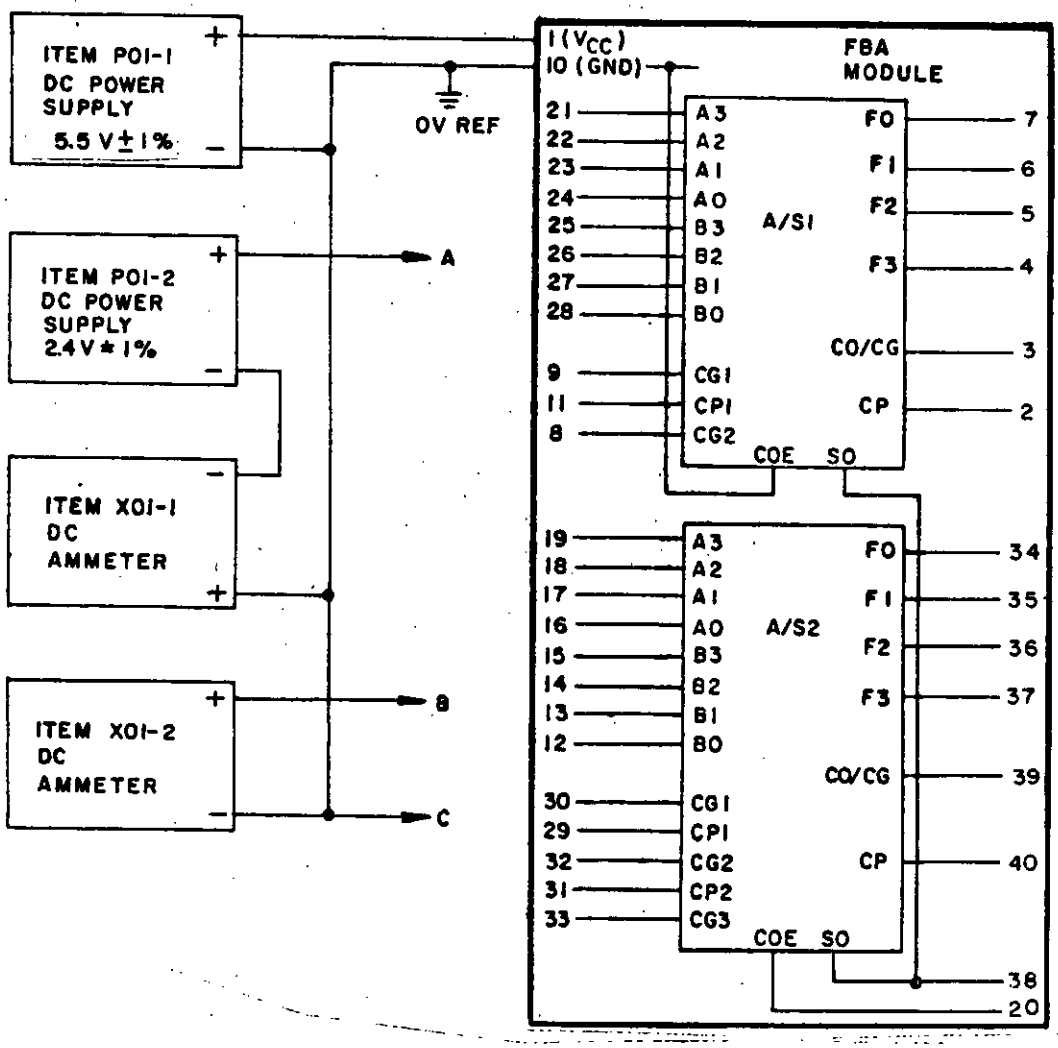
Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
H1	1A3	Input current, high level	21	Open	38, 25	0.05	60	µA	14
H2	1A2		22		38, 26				
H3	1A1		23		38, 27				
H4	1A0		24		38, 28				
H5	1B3		25		38, 21				
H6	1B2		26		38, 22				
H7	1B1		27		38, 23				
H8	1B0		28		38, 24				
H9	1CG1		9		11, 8				
H10	2A3		19		38, 15				
H11	2A2		18		38, 14				
H12	2A1		17		38, 13				
H13	2A0		16		38, 12				
H14	2B3		15		38, 19				
H15	2B2		14		38, 18				
H16	2B1		13		38, 17				
H17	2B0		12		38, 16				
H18	2CG1		30		32, 29, 31, 33				

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TABLE X. Input current (high-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
H19	1CG2	Input current, high level	8	Open	9	0.05	40	µA	14
H20	2CG2		32		29,30,31,33				
H21	SO		38		16,17,18, 19,21,22, 23,24				
H22	2COE		20		8		20		
H23	1CP1		11		9				
H24	2CP1		29		30,31,32,33				
H25	2CP2		31		29,30,32,33				
H26	2CG3		33		29,30,31,32				

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. With point A disconnected, item X01-1 should indicate zero current on the 1 µA range.

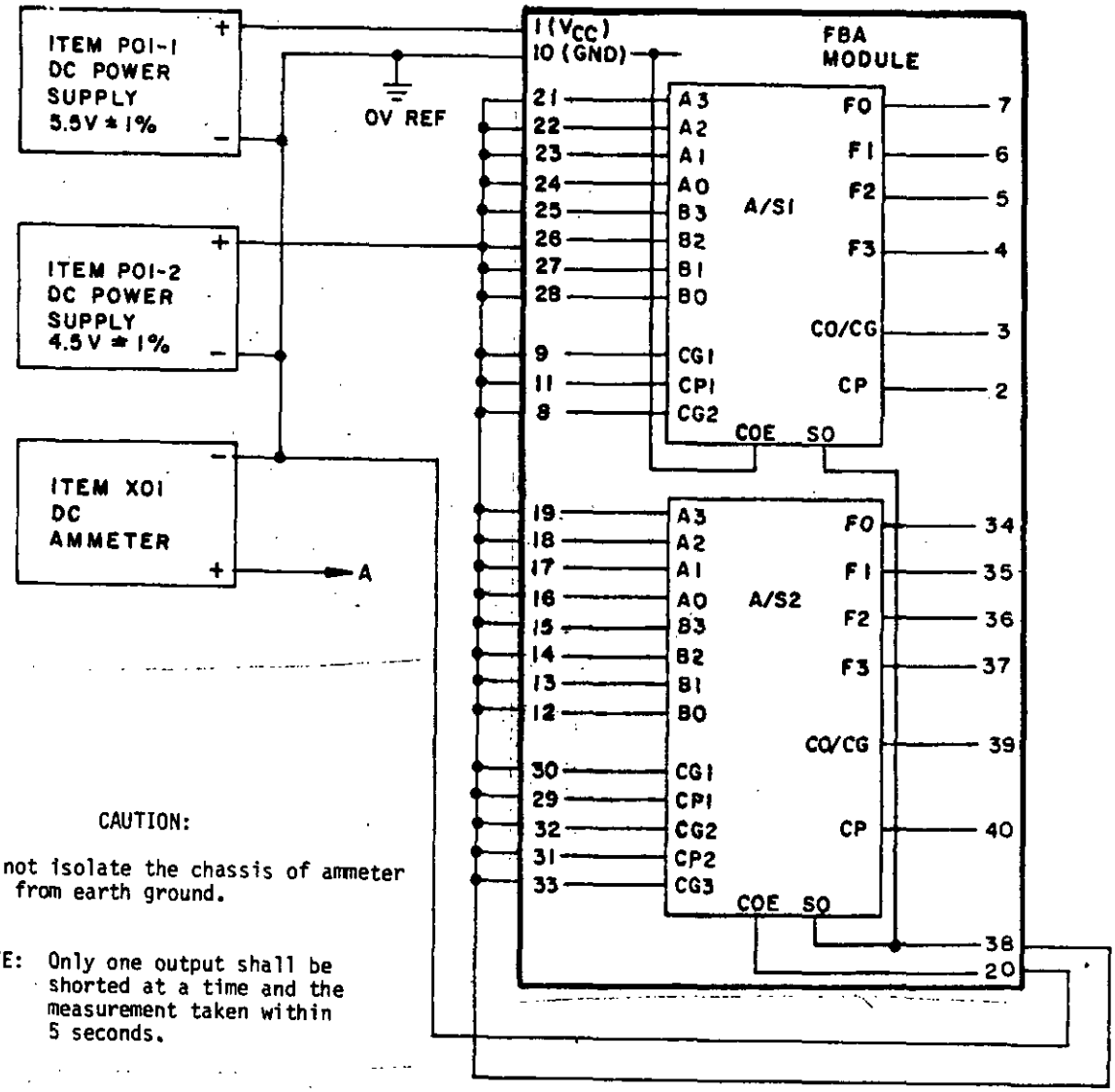
FIGURE 14. Input current (low-level and high-level) test circuit for tables IX and X.

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TABLE XI. Output short circuit current test.

Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
J1	1F0	Output short circuit current	7	10	40	mA	15
J2	1F1		6				
J3	1F2		5				
J4	1F3		4				
J5	2F0		34				
J6	2F1		35				
J7	2F2		36				
J8	2F3		37				
J9	1CO/CG		3				
J10	1CP		2				
J11	2CO/CG		39				
J12	2CP		40				

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTE: Only one output shall be shorted at a time and the measurement taken within 5 seconds.

FIGURE 15. Output short circuit current test circuit for table XI.

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TABLE XII. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
K	Power supply current	5	74	mA	16

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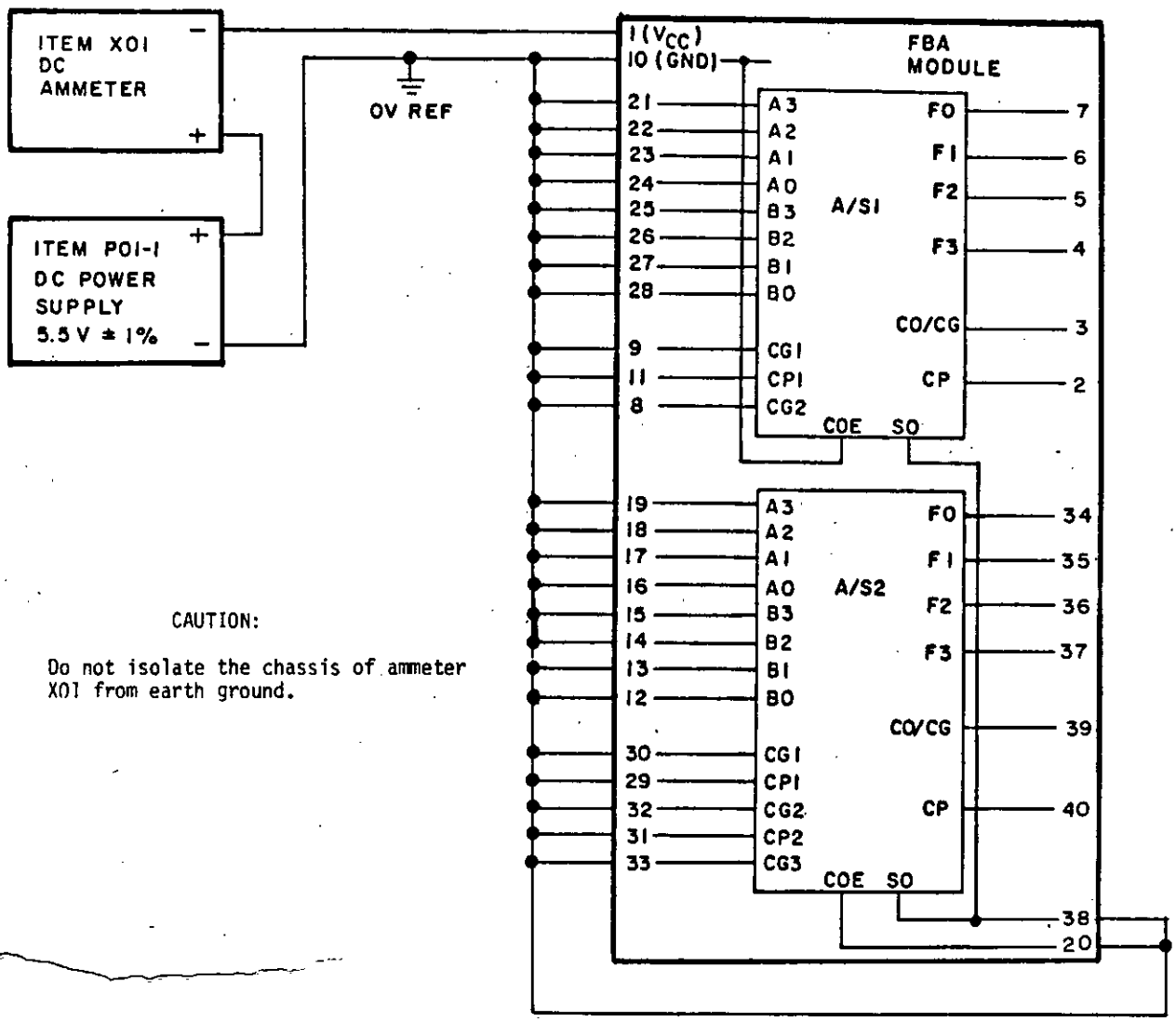


FIGURE 16. Power supply current test circuit for table XII.

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TABLE XIII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1F0 Transition time (t_{TLH}), C_L = 40 pF	28	7	38,9,11,26,27	21,22,23,24,25	2	45	ns	17,18
L2	1F1		6						
L3	1F2		5						
L4	1F3		4						
L5	1CG/C0		3	38,9,11,25,26,27	21,22,23,24				
L6	1CP		2	38,9,11,24,25, 26,27	21,22,23				
L7	2F0	12	34	38,29,30,13,14	15,16,17,18,19				
L8	2F1		35						
L9	2F2		36						
L10	2F3		37						
L11	2CG/C0		39	38,29,30,13,14,15	16,17,18,19,20				
L12	2CP		40	38,29,30,13,14, 15,16	17,18,19,20				

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TABLE XIII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L13	1F0 Transition time (t_{TLH}), C_L = 140 pF	28	7	38,9,11,26,27	21,22,23,24,25	4	60	ns	17,18
L14	1F1		6						
L15	1F2		5						
L16	1F3		4						
L17	1CG/C0		3	38,9,11,25,26,27	21,22,23,24				
L18	1CP		2	38,9,11,24,25, 26,27	21,22,23				
L19	2F0	12	34	38,29,30,13,14	15,16,17,18,19				
L20	2F1		35						
L21	2F2		36						
L22	2F3		37						
L23	2CG/C0		39	38,29,30,13,14,15	16,17,18,19,20				
L24	2CP		40	38,29,30,13,14, 15,16	17,18,19,20				

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TABLE XIV. Transition time (t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1FO Transition time (t_{THL}), C_L = 40 pF	28	7	38,9,11,26,27	21,22,23,24,25	2	15	ns	17,18
M2	1F1		6						
M3	1F2		5						
M4	1F3		4						
M5	1CG/CO		3	38,9,11,25,26,27	21,22,23,24				
M6	1CP		2	38,9,11,24,25, 26,27	21,22,23				
M7	2FO	12	34	38,29,30,13,14	15,16,17,18,19				
M8	2F1		35						
M9	2F2		36						
M10	2F3		37						
M11	2CG/CO		39	38,29,30,13,14,15	16,17,18,19,20				
M12	2CP		40	38,29,30,13,14,15 16	17,18,19,20				
M13	1FO Transition time (t_{THL}), C_L = 140 pF	28	7	38,9,11,26,27	21,22,23,24,25		30		

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TABLE XIV. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M14	1F1 Transition time (t_{THL}), C_L = 140 pF	28	6	38,9,11,26,27	21,22,23,24,25	2	30	ns	17,18
M15	1F2		5						
M16	1F3		4						
M17	1CG/CO		3	38,9,11,25,26,27	21,22,23,24				
M18	1CP		2	38,9,11,24,25, 26,27	21,22,23				
M19	2F0	12	34	38,29,30,13,14	15,16,17,18,19				
M20	2F1		35						
M21	2F2		36						
M22	2F3		37						
M23	2CG/CO		39	38,29,30,13,14,15	16,17,18,19,20				
M24	2CP		40	38,29,30,13,14, 15,16	17,18,19,20				

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TABLE XV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	B0 to F3, Add — Propagation delay time (t_{PLH}), C_L = 40 pF	28	4	38,9,11,26,27	21,22,23,24,25	10	101	ns	17,18
N2	B0 to F3, Subtract		4	9,11,25	38,21,22,23,24,26,27	12	101		
N3	B0 to C0/CG, Add		3	38,9,11,25,26,27	21,22,23,24	7	68		
N4	B0 to C0/CG, Subtract		3	9,11	38,21,22,23,24,25,26,27	9	65		
N5	B0 to CP, Add		2	38,9,11,24,25,26,27	21,22,23	7	68		
N6	B0 to F0, Add		7	38,9,11,25,26,27	21,22,23,24	10	101		
N7	B1 to F1, Add	27	6	38,9,11,25,26,28					
N8	B2 to F2, Add	26	5	38,9,11,25,27,28					
N9	B3 to F3, Add	25	4	38,9,11,26,27,28					
N10	A0 to F0, Add	24	7	38,9,11,21,22,23	25,26,27,28				
N11	A1 to F1, Add	23	6	38,9,11,21,22,24	25,26,27,28				
N12	A2 to F2, Add	22	5	38,9,11,21,23,24	25,26,27,28				

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N13	A3 to F3, Add — Propagation delay time (t_{PLH}), C_L = 40 pF	21	4	38,9,11,22,23,24	25,26,27,28	10	101	ns	17,18
N14	CG1 to F3, Add	9	4	38,11,21,22,23, 24,25	26,27,28	10	94		
N15	B0 to F3, Add	12	37	38,29,30,13,14	15,16,17,18,19	10	101		
N16	B0 to F3, Subtract		37	15,29,30	38,13,14,16,17, 18,19	12	101		
N17	B0 to C0/CG, Add		39	38,29,30,13,14,15	16,17,18,19,20	7	68		
N18	B0 to C0/CG, Subtract		39	29,30	38,13,14,15,16, 17,18,19,20	9	65		
N19	B0 to CP, Add		40	38,29,30,13,14, 15,16	17,18,19,20	7	68		
N20	B0 to F0, Add		34	38,13,14,15,29, 30	16,17,18,19	10	101		
N21	B1 to F1, Add	13	35	38,12,14,15,29, 30					
N22	B2 to F2, Add	14	36	38,12,13,15,29, 30					
N23	B3 to F3, Add	15	37	38,12,13,14,29, 30					
N24	A0 to F0, Add	16	34	38,17,18,19,29, 30	12,13,14,15				
N25	A1 to F1, Add	17	35	38,16,18,19,29, 30	12,13,14,15				

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N26	A2 to F2, Add — Propagation delay time (t_{PLH}), C_L = 40 pF	18	36	38,16,17,19,29,30	12,13,14,15	10	101	ns	17,18
N27	A3 to F3, Add	19	37	38,16,17,18,29, 30	12,13,14,15	10	101		
N28	CG3 to F3, Add	33	37	38,30,32,15,16, 17,18,19	12,13,14,29,31	10	94		
N29	CG2 to CO/CG, Add	32	39	38,30,31,16,17, 18,19,20	12,13,14,15,29	11	58		
N30	CG1 to CO/CG, Add	30	39	38,31,32,16,17, 18,19,20	12,13,14,15,29	11	58		
N31	CG3 to CO/CG, Add	33	39	38,30,32,16,17, 18,19,20	12,13,14,15,29,31	11	58		
N32	B0 to F3, Add — Propagation delay time (t_{PLH}), C_L = 140 pF	28	4	38,9,11,26,27	21,22,23,24,25	10	121		
N33	B0 to F3, Subtract		4	9,11,25	38,21,22,23,24, 26,27	12	121		
N34	B0 to CO/CG, Add		3	38,9,11,25,26,27	21,22,23,24	10	88		
N35	B0 to CO/CG, Subtract		3	9,11	38,21,22,23,24,25, 26,27		85		
N36	B0 to CP, Add		2	38,9,11,24,25, 26,27	21,22,23		88		
N37	B0 to F0, Add		7	38,9,11,25,26,27	21,22,23,24		121		

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N38	B1 to F1, Add — Propagation delay time (t_{PLH}), C_L = 140 pF	27	6	38,9,11,25,26,28	21,22,23,24	10	121	ns	17,18
N39	B2 to F2, Add	26	5	38,9,11,25,27,28					
N40	B3 to F3, Add	25	4	38,9,11,26,27,28					
N41	A0 to F0, Add	24	7	38,9,11,21,22,23	25,26,27,28				
N42	A1 to F1, Add	23	6	38,9,11,21,22,24					
N43	A2 to F2, Add	22	5	38,9,11,21,23,24					
N44	A3 to F3, Add	21	4	38,9,11,22,23,24					
N45	CG1 to F3, Add	9	4	38,11,21,22,23, 24,25	11,26,27,28		114		
N46	B0 to F3, Add	12	37	38,29,30,13,14	15,16,17,18,19		121		
N47	B0 to F3, Subtract		37	15,29,30	38,13,14,16,17, 18,19	12	121		
N48	B0 to C0/CG, Add		39	38,29,30,13,14, 15	16,17,18,19,20	10	88		
N49	B0 to C0/CG, Subtract		39	29,30	38,13,14,15,16,17, 18,19,20		85		
N50	B0 to CP, Add		40	38,29,30,13,14, 15,16	17,18,19,20		88		
N51	B0 to F0, Add		34	38,13,14,15,29, 30	16,17,18,19		121		
N52	B1 to F1, Add	13	35	38,12,14,15,29, 30	16,17,18,19		121		

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N53	B2 to F2, Add — Propagation delay time (t_{PLH}), C_L = 140 pF	14	36	38,12,13,15,29,30	16,17,18,19	10	121	ns	17,18
N54	B3 to F3, Add	15	37	38,12,13,14,29,30	16,17,18,19				
N55	A0 to F0, Add	16	34	38,17,18,19,29,30	12,13,14,15				
N56	A1 to F1, Add	17	35	38,16,18,19,29,30					
N57	A2 to F2, Add	18	36	38,16,17,19,29,30					
N58	A3 to F3, Add	19	37	38,16,17,18,29,30					
N59	CG3 to F3, Add	33	37	38,30,32,15,16,17,18,19	12,13,14,29,31		114		
N60	CG2 to CO/CG, Add	32	39	38,30,31,16,17,18,19,20	12,13,14,15,29	11	78		
N61	CG1 to CO/CG, Add	30	39	38,31,32,16,17,18,19,20	12,13,14,15,29	11	78		
N62	CG3 to CO/CG, Add	33	39	38,30,32,16,17,18,19,20	12,13,14,15,29,31	11	78		

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TABLE XVI. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	B0 to F3, Add Propagation delay time (t_{PHL}), C_L = 40 pF	28	4	38,9,11,26,27	21,22,23,24,25	10	81	ns	17,18
P2	B0 to F3, Subtract		4	9,11,25	38,21,22,23,24, 26,27	12	100		
P3	B0 to C0/CG, Add		3	38,9,11,25,26,27	21,22,23,24,	7	48		
P4	B0 to C0/CG, Subtract		3	9,11	38,21,22,23,24, 25,26,27	9	67		
P5	B0 to CP, Add		2	38,9,11,24,25, 26,27	21,22,23	7	48		
P6	B0 to F0, Add		7	38,9,11,25,26, 27	21,22,23,24	10	81		
P7	B1 to F1, Add	27	6	38,9,11,25,26, 28	21,22,23,24				
P8	B2 to F2, Add	26	5	38,9,11,25,27, 28					
P9	B3 to F3, Add	25	4	38,9,11,26,27, 28					
P10	A0 to F0, Add	24	7	38,9,11,21,22, 23	25,26,27,28				
P11	A1 to F1, Add	23	6	38,9,11,21,22, 24					
P12	A2 to F2, Add	22	5	38,9,11,21,23, 24					
P13	A3 to F3, Add	21	4	38,9,11,22,23, 24					
P14	CG1 to F3, Add	9	4	38,11,21,22, 23,24,25	26,27,28		82		

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TABLE XVI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P15	B0 to F3, Add Propagation delay time (t_{PHL}), C_L = 40 pF	12	37	38,29,30,13,14	15,16,17,18,19	10	81	ns	17,18
P16	B0 to F3, Subtract		37	15,29,30	38,13,14,16,17, 18,19	12	100		
P17	B0 to C0/CG, Add		39	38,29,30,13,14, 15	16,17,18,19,20	7	48		
P18	B0 to C0/CG, Subtract		39	29,30	38,13,14,15,16, 17,18,19,20	9	67		
P19	B0 to CP, Add		40	38,29,30,13,14, 15,16	17,18,19,20	7	48		
P20	B0 to F0, Add		34	38,13,14,15,29, 30	16,17,18,19	10	81		
P21	B1 to F1, Add	13	35	38,12,14,15,29, 30					
P22	B2 to F2, Add	14	36	38,12,13,15,29, 30					
P23	B3 to F3, Add	15	37	38,12,13,14,29, 30					
P24	A0 to F0, Add	16	34	38,17,18,19,29, 30	12,13,14,15				
P25	A1 to F1, Add	17	35	38,16,18,19,29, 30	12,13,14,15				

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TABLE XVI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P26	A2 to F2, Add Propagation delay time (t_{PHL}), C_L = 40 pF	18	36	38,17,19,29,30, 16	12,13,14,15	10	81	ns	17,18
P27	A3 to F3, Add	19	37	38,16,17,18,29, 30	12,13,14,15	10	81		
P28	CG3 to F3, Add	33	37	38,30,32,15,16, 17,18,19	12,13,14,29,31	10	82		
P29	CG2 to CO/CG, Add	32	39	38,30,31,16,17, 18,19,20	12,13,14,15,29	7	50		
P30	CG1 to CO/CG, Add	30	39	38,31,32,16,17, 18,19,20	12,13,14,15,29	7	50		
P31	CG3 to CO/CG, Add	33	39	38,30,32,16,17, 18,19,20	12,13,14,15,29,31	7	50		
P32	BO to F3, Add Propagation delay time (t_{PHL}), C_L = 140 pF	28	4	38,9,11,26,27	21,22,23,24,25	10	101		
P33	BO to F3, Subtract		4	9,11,25	38,21,22,23,24, 26,27	12	120		
P34	BO to CO/CG, Add		3	38,9,11,25,26,27	21,22,23,24	10	68		
P35	BO to CO/CG, Subtract		3	9,11	38,21,22,23,24,25, 26,27		87		
P36	BO to CP, Add		2	38,9,11,24,25, 26,27	21,22,23		68		
P37	BO to FO, Add		7	38,9,11,25,26, 27	21,22,23,24		101		

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TABLE XVI. Propagation delay time (t_{PHL}) test - Continued.

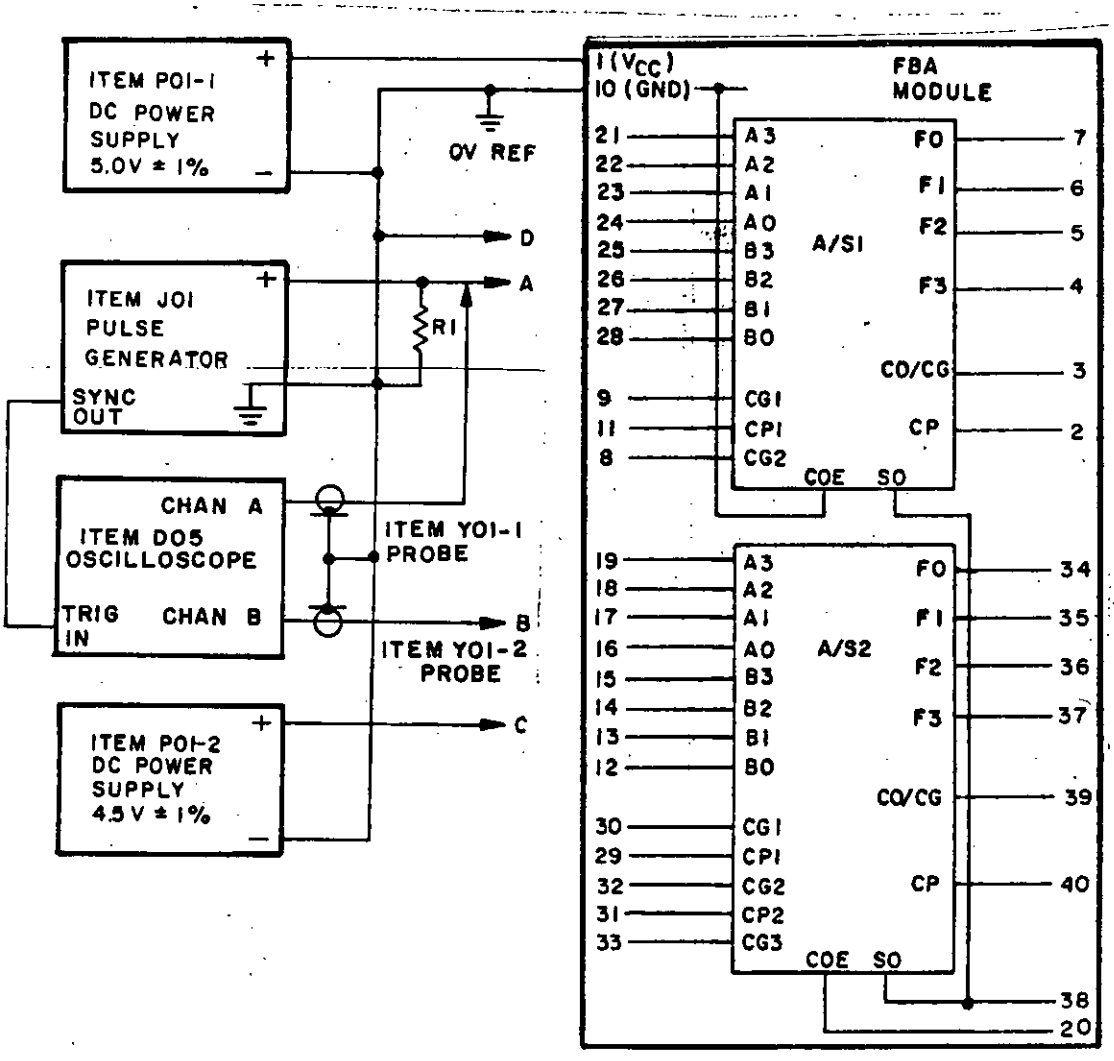
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P38	B1 to F1, Add Propagation delay time (t_{PHL}), C_L = 140 pF	27	6	38,9,11,25,26,28	21,22,23,24	10	101	ns	17,18
P39	B2 to F2, Add	26	5	38,9,11,25,27,28					
P40	B3 to F3, Add	25	4	38,9,11,26,27,28					
P41	A0 to F0, Add	24	7	38,9,11,21,22,23	25,26,27,28				
P42	A1 to F1, Add	23	6	38,9,11,21,22,24					
P43	A2 to F2, Add	22	5	38,9,11,21,23,24					
P44	A3 to F3, Add	21	4	38,9,11,22,23,24					
P45	CG1 to F3, Add	9	4	38,11,21,22,23, 24,25	26,27,28		102		
P46	B0 to F3, Add	12	37	38,29,30,13,14	15,16,17,18,19		101		
P47	B0 to F3, Subtract		37	15,29,30	38,13,14,16,17, 18,19	12	120		
P48	B0 to C0/CG, Add		39	38,29,30,13,14, 15	16,17,18,19,20	10	68		
P49	B0 to C0/CG, Subtract		39	29,30	38,13,14,15,16, 17,18,19,20	10	87		
P50	B0 to CP, Add		40	38.29,30,13,14, 15,16	17,18,19,20	10	68		

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TABLE XVI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P51	B0 to F0, Add — Propagation delay time (t_{PHL}), C_L = 140 pF	12	34	38,13,14,15,29,30	16,17,18,19	10	101	ns	17,18
P52	B1 to F1, Add	13	35	38,12,14,15,29,30					
P53	B2 to F2, Add	14	36	38,12,13,15,29,30					
P54	B3 to F3, Add	15	37	38,12,13,14,29,30					
P55	A0 to F0, Add	16	34	38,17,18,19,29,30	12,13,14,15				
P56	A1 to F1, Add	17	35	38,16,18,19,29,30					
P57	A2 to F2, Add	18	36	38,17,19,29,30,16					
P58	A3 to F3, Add	19	37	38,16,17,18,29,30					
P59	CG3 to F3, Add	33	37	38,30,32,15,16,17,18,19	12,13,14,29,31		102		
P60	CG2 to CO/CG, Add	32	39	38,30,31,16,17,18,19,20	12,13,14,15,29		70		
P61	CG1 to CO/CG, Add	30	39	38,31,32,16,17,18,19,20	12,13,14,15,29		70		
P62	CG3 to CO/CG, Add	33	39	38,30,32,16,17,18,19,20	12,13,14,15,29,31		70		

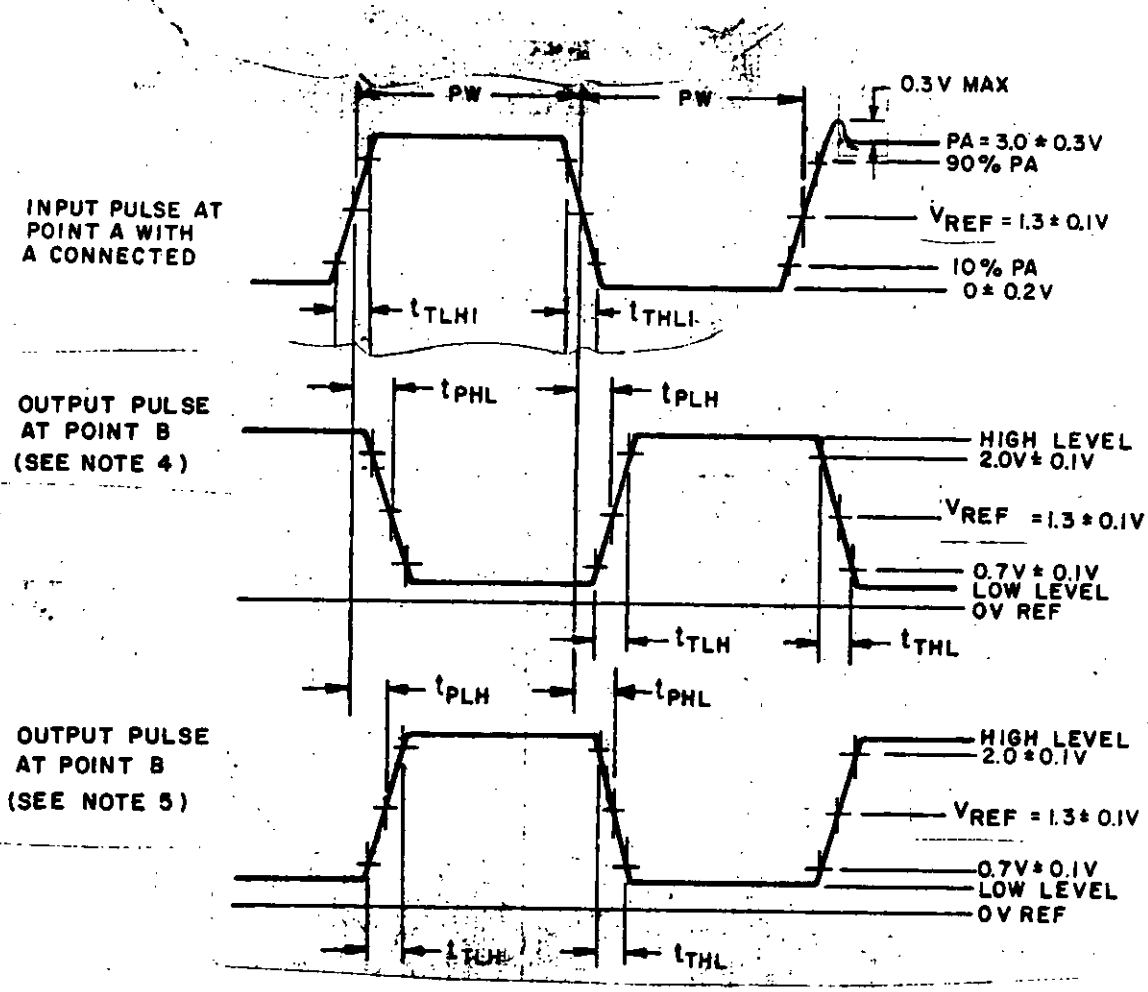
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load all outputs with load as specified on figure 12.

FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XIII, XIV, XV and XVI.

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NOTES:
1. Input pulse characteristics:
   PW = 500 ±50 ns.
   $t_{THL1} = t_{TLH1} = 10 \pm 2$ ns.
2. All voltage levels are referenced to ground.
3. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.3 V.
4. Output pulse for test conditions L1-L24, M1-M24, N2, N4, N6-N13, N16, N18, N21-N27, N32, N34, N36-N46, N48, N50-N62, P1, P3, P5-P15, P17, P19-P32, P34, P36-P46, P48 and P50-P62.
5. Output pulse for test conditions N1, N3, N5, N14, N15, N17, N19, N20, N28-N31, N33, N35, N47, N49, P2, P4, P16, P18, P33, P35, P47, and P49.

FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIII, XIV, XV and XVI.

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TABLE XVII. Output crosstalk (low-level) test.

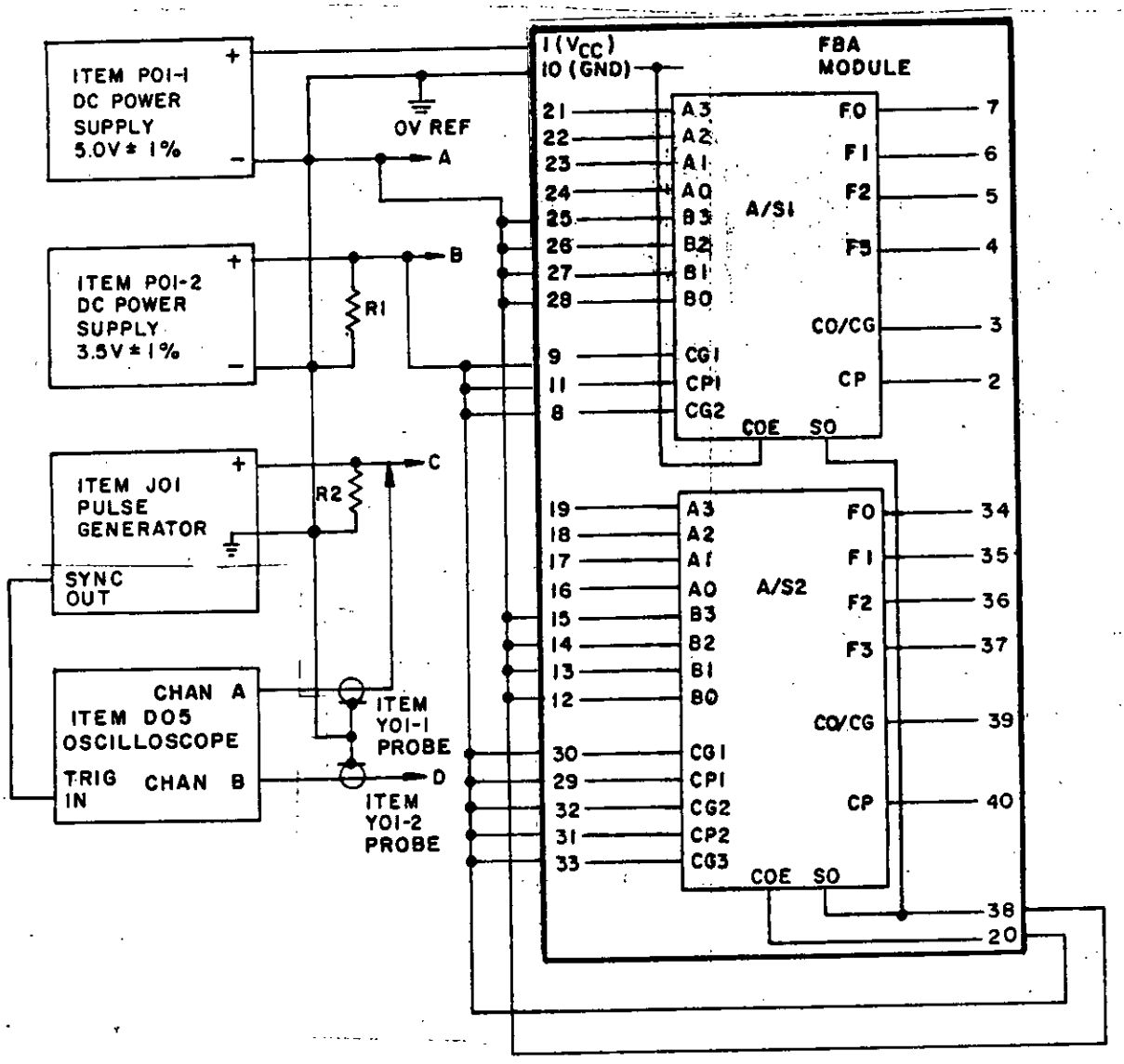
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1F0 Output crosstalk, low level	24	Open	16,17,18,19, 21,22,23	7	0	400	mV	19,20
R2	1F1	23		16,17,18,19, 21,22,24	6				
R3	1F2	22		16,17,18,19, 21,23,24	5				
R4	1F3	21		16,17,18,19, 22,23,24	4				
R5	2F0	16		17,18,19,21, 22,23,24	34				
R6	2F1	17		16,18,19,21, 22,23,24	35				
R7	2F2	18		16,17,19,21, 22,23,24	36				
R8	2F3	19		16,17,18,21, 22,23,24	37				

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TABLE XVIII. Output crosstalk (high-level) test.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1F0	Output crosstalk, high level	Open	24	16,17,18,19, 21,22,23	7	0	400	mV	19,20
S2	1F1			23	16,17,18,19, 21,22,24	6				
S3	1F2			22	16,17,18,19, 21,23,24	5				
S4	1F3			21	16,17,18,19, 22,23,24	4				
S5	2F0			16	17,18,19,21, 22,23,24	34				
S6	2F1			17	16,18,19,21, 22,23,24	35				
S7	2F2			18	16,17,19,21, 22,23,24	36				
S8	2F3			19	16,17,18,21, 22,23,24	37				

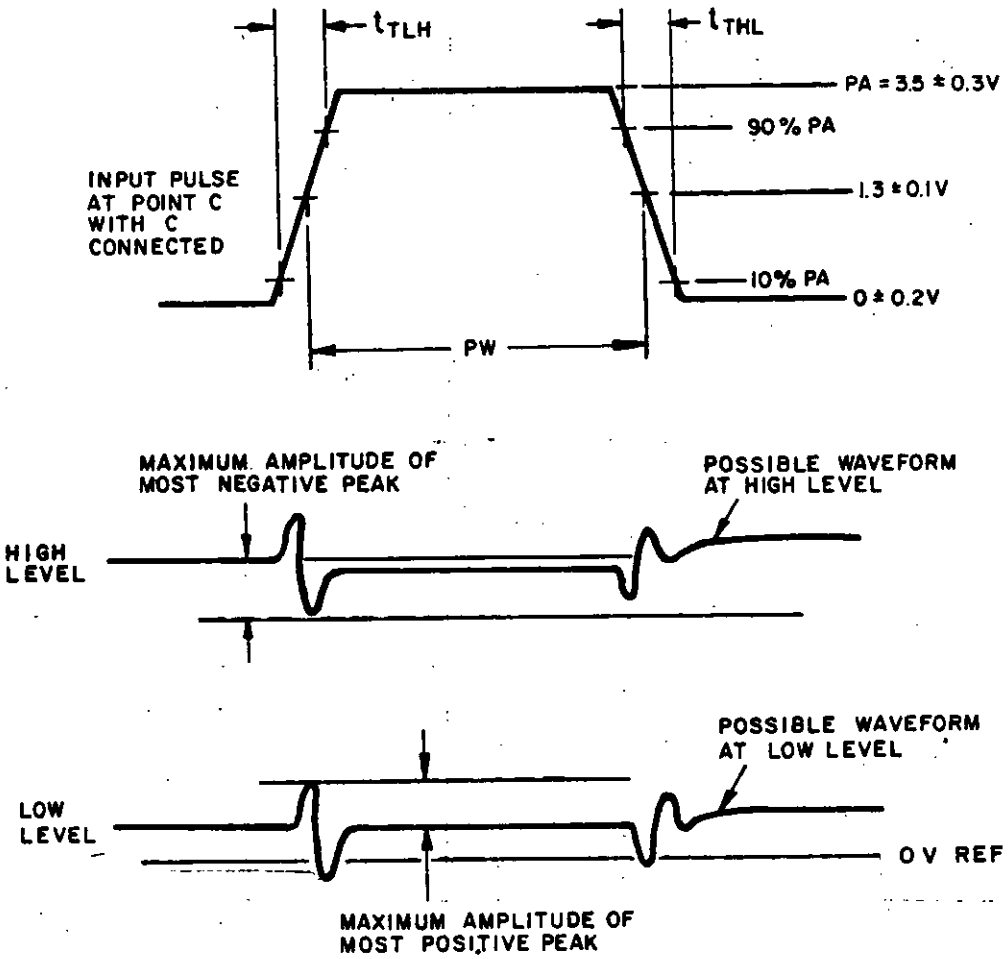
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load each output with the load as specified on figure 12.

FIGURE 19. Output crosstalk (low-level and high-level) test circuit for tables XVII and XVIII.

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NOTES:
1. Input pulse parameters:
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL} = 20 \pm 2$ ns.
   PRR = 1 MHz ±10%.
2. All input voltage levels are referenced to ground.

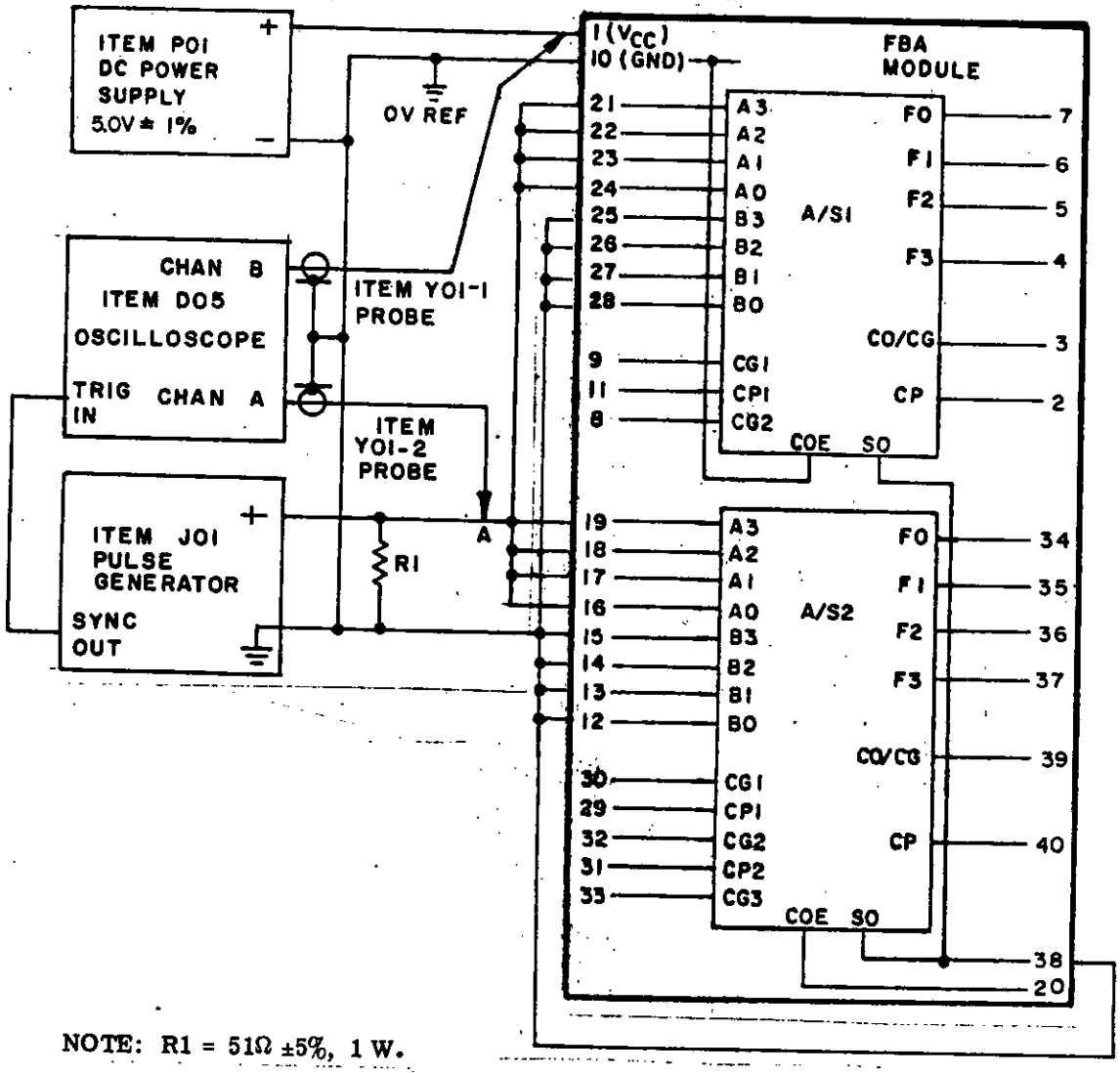
FIGURE 20. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVII and XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient amplitude	0	300	mV	21,22

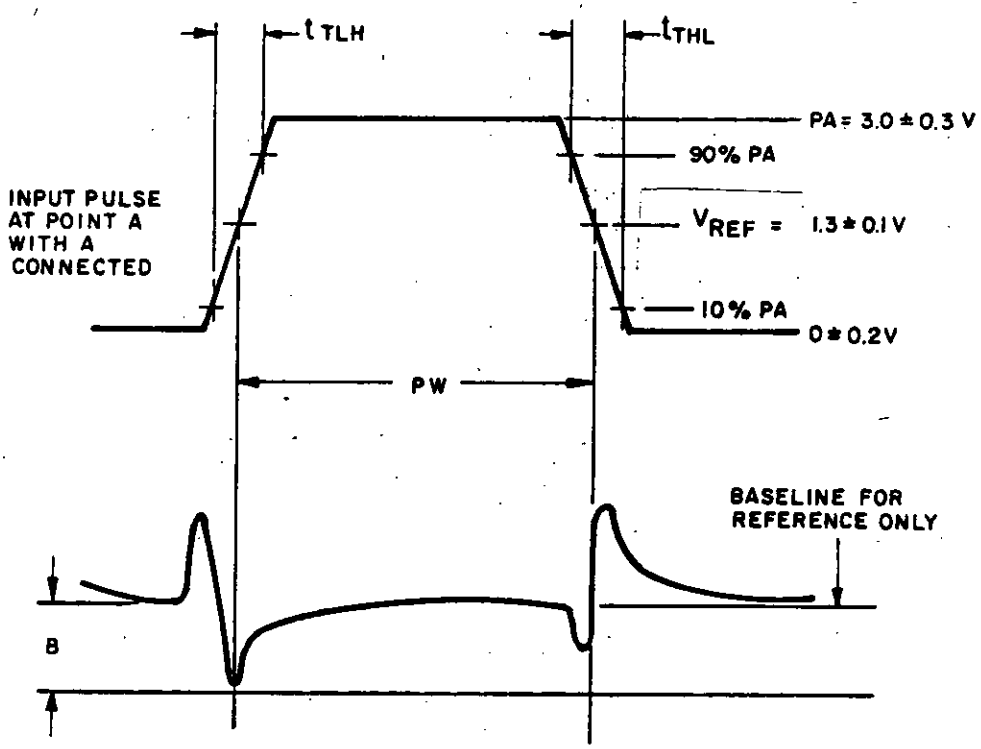
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 21. Power supply transient amplitude test circuit for table XIX.

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NOTES:
1. Input pulse parameters:
   PW = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 20 ±2 ns (at point A).
   PRR = 1 MHz ±10%.
2. B = Measurement from the baseline to the most negative excursion of the waveform.
3. All input voltage levels are referenced to ground.

FIGURE 22. Power supply transient amplitude test circuit waveforms for table XIX.

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TABLE XX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U	Power supply filter capacitance	Open	1	1.65	2.75	µF	23

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TABLE XXI. Input capacitance test.

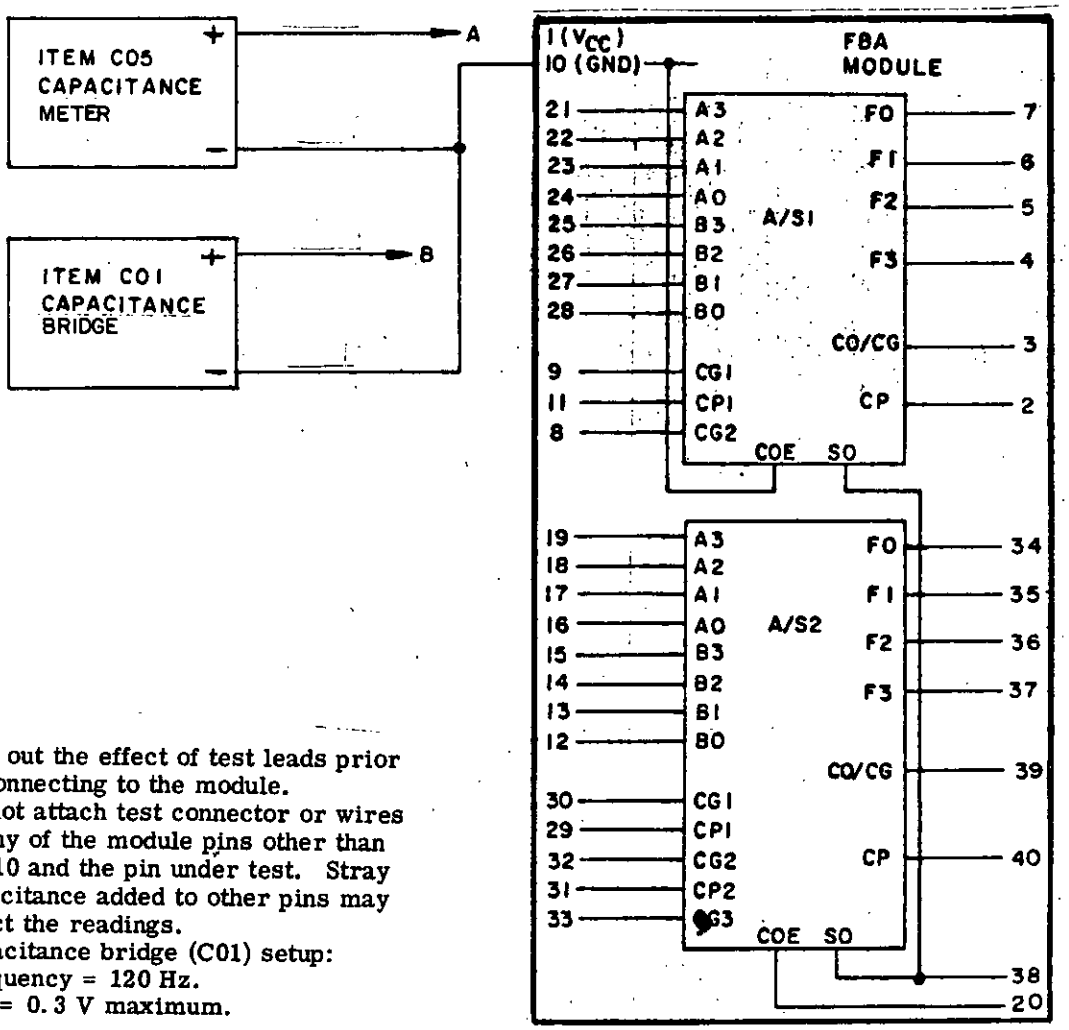
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	1A3	21	Open	2	15	pF	23
V2	1A2	22					
V3	1A1	23					
V4	1A0	24					
V5	1B3	25					
V6	1B2	26					
V7	1B1	27					
V8	1B0	28					
V9	2A3	19					
V10	2A2	18					
V11	2A1	17					
V12	2A0	16					
V13	2B3	15					
V14	2B2	14					
V15	2B1	13					
V16	2B0	12					
V17	1CG1	9					
V18	1CG2	8					
V19	1CP1	11					

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TABLE XXI. Input capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
V20	2CP1	Input capacitance	29	Open	2	15	pF	23
V21	2CP2	↓	31	↓	↓	↓	↓	↓
V22	2CG1		30					
V23	2CG2		32					
V24	2CG3		33					
V25	2COE		20					
V26	SO		38					

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NOTES:
1. Null out the effect of test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 23. Power supply filter capacitance and input capacitance test circuit for tables XX and XXI.

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TABLE XXII. Functional test.

Test cond.	Connect			Output logic states module output pins												Initial limits 25° ±5°C ambient		Unit	Test fig.
	A to pin	B to pin	C to pin	7	6	5	4	3	2	34	35	36	37	39	40	Min	Max		
W1	8,11,12,16,18,19,21, 22,24,28,29,31,32, 33,38	9,13,14,15,17,20, 23,25,26,27,30	2/	L	H	H	H	L	H	L	H	H	H	L	H	1/	1/	1/	24
W2	8,11,16,17,18,19,21, 22,23,24,29,31,32, 33	9,12,13,14,15,20, 25,26,27,28,30,38		L	H	H	H	H	H	L	H	H	H	H	H				
W3	8,9,11,16,17,18,19, 21,22,23,24,29,30, 31,32,33,38	12,13,14,15,20,25, 26,27,28		L	L	L	L	H	L	L	L	L	L	H	L				
W4	8,9,12,13,14,15,25, 26,27,28,30,31,32, 33	11,16,17,18,19,20, 21,22,23,24,29,38		H	L	L	L	L	L	H	L	L	L	L	L				
W5	9,12,13,14,15,25, 26,27,28,30,31,33	8,11,16,17,18,19, 20,21,22,23,24,29, 32,38		L	L	L	L	L	L	L	L	L	L	L	L				
W6	8,11,15,19,21,25,29, 31,32,33,38	9,12,13,14,16,17, 18,20,22,23,24,26, 27,28,30		L	L	L	L	H	H	L	L	L	L	H	H				
W7	9,11,12,15,17,19,21, 23,25,28,29,30,31, 33,38	8,13,14,16,18,20, 22,24,26,27,32		L	L	H	L	H	H	L	L	H	L	H	H				
W8	8,9,11,13,15,17,23, 25,27,29,30,31,32, 33,38	12,14,16,18 19,20, 21,22,24,26,28		H	L	H	H	L	H	H	L	H	H	L	H				
W9	12,15,16,17,18,22, 23,24,25,28,31,33,38	8,9,11,13,14,19,20, 21,26,27,29,30,32		L	L	L	L	H	H	L	L	L	L	H	H				
W10	8,12,13,14,15,16,17, 18,19,21,22,23,24, 25,26,27,28,31,32,33	9,11,20,29,30,38		H	H	H	H	H	L	H	H	H	H	H	L				

See footnotes at end of table.

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TABLE XXII. Functional test - Continued.

Test cond.	Connect			Output logic states module output pins												Initial limits 25° ±5°C ambient		Unit	Test fig.
	A to pin	B to pin	C to pin	7	6	5	4	3	2	34	35	36	37	39	40	Min	Max		
W11	8,11,13,17,19,21,22, 23,27,29,31,32,33, 38,18	9,12,14,15,16,20, 24,25,26,28,30	2/	L	L	L	L	H	H	L	L	L	L	H	H	1/	1/	1/	24
W12	11,14,17,18,22,23, 26,29,31,33,38	8,9,12,13,15,16,19, 20,21,24,25,27,28, 30,32		L	H	L	H	L	H	L	H	L	H	L	H				
W13	8,9,11,14,18,19,21, 22,26,29,30,31,32, 33,38	12,13,15,16,17,20, 23,24,25,27,28		H	L	L	L	H	H	H	L	L	L	H	H				
W14	31,33	8,9,11,12,13,14,15, 16,17,18,19,20,21, 22,23,24,25,26,27, 28,29,30,32,38		H	H	H	H	H	L	H	H	H	H	H	L				
W15	8,9,11,17,18,19,21, 22,23,29,30,31,32, 33,38	12,13,14,15,16,20, 24,25,26,27,28		H	H	H	H	L	L	H	H	H	H	L	L				
W16	16,17,18,19,20,30, 32,38	12,13,14,15,29,31, 33		-	-	-	-	-	-	H	H	H	H	L	L				
W17	13,15,17,19,20,30, 32,33,38	12,14,16,18,29,31,		-	-	-	-	-	-	H	L	H	L	H	H				
W18	12,14,16,18,20,30, 31,32,38	13,15,17,19,29,33		-	-	-	-	-	-	H	H	L	H	L	H				
W19	16,17,18,19,20,29, 30,38	12,13,14,15,31,32, 33		-	-	-	-	-	-	L	L	L	L	H	L				
W20	12,16,17,18,19,20, 30,32,38	13,14,15,29,31,33		-	-	-	-	-	-	L	L	L	L	H	H				

See footnotes at end of table.

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TABLE XXII. Functional test - Continued.

Test cond.	Connect			Output logic states module output pins												Initial limits 25° ±5°C ambient		Unit	Test fig.
	A to pin	B to pin	C to pin	7	6	5	4	3	2	34	35	36	37	39	40	Min	Max		
W21	13,16,17,18,19,20, 30,32,38	12,14,15,29,31,33	2/	-	-	-	-	-	-	H	L	L	L	H	H	1/	1/	1/	24
W22	14,16,17,18,19,20, 30,32,38	12,13,15,29,31,33		-	-	-	-	-	-	H	H	L	L	H	H				
W23	15,16,17,18,19,20, 30,32,38	12,13,14,29,31,33		-	-	-	-	-	-	H	H	H	L	H	H				
W24	9,11,14,15,19,21,25, 26,29,30,31,33,38	8,12,13,16,17,18, 20,22,23,24,27,28, 32		H	L	H	L	H	H	H	L	H	L	H	H				
W25	8,11,13,27,29,31,32, 33,38	9,12,14,15,16,17, 18,19,20,21,22,23, 24,25,26,28,30		L	H	L	L	L	L	L	H	L	L	L	L				

1/ Output logic states
L = Low voltage level = 0.0 to 0.3 V
H = High voltage level = 2.4 to 4.5 V

2/ For test conditions W1-W25, verification of the output logic state of footnote 1/ is obtained by connecting item V03, point C, to each output pin in succession after input conditions have been accomplished.

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